"A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains"

MAJOR PROJECT REPORT

Submitted in partial fulfillment of the requirements for the degree of
BACHELOR OF ENGINEERING
in Artificial Intelligence and Data Science

by

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Academic Year 2026–2027

Approval Sheet

This Major Project Report entitled "A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains" submitted by the students listed below is approved for the degree of Bachelor of Engineering in Artificial Intelligence and Data Science, Academic Year 2026–2027.

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CERTIFICATE

This is to certify that the project entitled "A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains" is a bonafide work carried out by Doctor UmerKhan JunaidKhan (TU8S2425008), Kerkar Suyash Satish (TU8S2425003), Esaf Shanum Shakil (TU8S2425004), and Patil Rutuja Sanjeev (TU8S2425006) in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Artificial Intelligence and Data Science at TPCT's Terna Engineering College, Nerul (W), Navi Mumbai, affiliated to Mumbai University, during the academic year 2026–2027.

To the best of our knowledge, this work has not been submitted to any other university or institution for the award of any other degree or diploma.

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Declaration

We declare that this written submission entitled "A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains" represents our own work based on our design, implementation, and analysis. Where ideas or text from external literature have been referenced, they have been duly cited in the references.

We also declare that we have adhered to academic honesty and ethical research practices. We understand that any unattributed reproduction of external material violates institutional policy and can evoke disciplinary action.

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Abstract

Agricultural supply chains in developing economies frequently suffer from post-harvest spoilage, fragmented cold-chain surveillance, and opaque middleman markups that leave smallholder farmers with a disproportionately low share of the final retail price. Although public and consortium blockchains offer immutable, auditable ledgers for product provenance, streaming raw IoT telemetry directly to smart contracts exposes the system to the 'Garbage In, Garbage Out' (GIGO) dilemma. When sensor telemetry suffers from thermal spikes, communication dropouts, drift, or replay attempts, committing such data directly onto the blockchain creates permanent false records, unnecessary disputes, and costly transaction gas expenditure.

In this project, we designed and implemented AgriChain, a unified decentralized architecture that integrates a simulated IoT telematics engine, an off-chain AI-based Trust Layer, modular smart contracts on the Polygon Amoy testnet, dual-engine relational persistence, and a responsive multi-role React 18 web interface. Simulated edge nodes model realistic temperature, humidity, and GPS waypoints across transit and warehousing phases. Before any reading reaches the blockchain, our off-chain AI Trust Layer validates it using a multi-stage pipeline: basic format checks, physical and temporal plausibility bounding, dynamic per-crop thermal envelopes, 11-dimensional feature extraction, unsupervised Isolation Forest anomaly scoring, and SHA-256 anti-replay verification. Readings that violate policy or exhibit anomalous signatures are routed into an append-only SHA-256 Quarantine Hash Chain, preserving a tamper-evident audit record without ledger pollution or wasted gas.

Valid readings are batched and committed to modular Solidity smart contracts (ProductRegistry, CustodyTransfer, ColdChainMonitor, PolicyConfig). A bitwise packed struct representation stores complete environmental condition records in a single 256-bit EVM storage slot, achieving a 52.71% gas reduction under batched updates (N=20). Off-chain quality certificates are pinned to IPFS with immutable Content Identifiers (CIDs) anchored on-chain. Consumers can scan a batch QR code to inspect the complete journey timeline, freshness ratings, verified reviews, and a cumulative price trail that demonstrates an equitable 50.0% farmer revenue share. Evaluated across 1,000 simulated telemetry cycles across seven realistic cold-chain failure profiles, the AI Trust Layer achieved an F1-score of 97.22% with 100% recall, providing an accountable and practical solution for agricultural supply chain transparency.

Keywords: Agricultural Supply Chain, Blockchain Traceability, Simulated IoT Telematics, AI Trust Layer, Isolation Forest Anomaly Detection, Modular Smart Contracts, Gas Optimization, Fair Price Trail, Quarantine Hash Chain.

Table of Contents

Approval Sheet ii
Certificate iii
Declaration iv
Acknowledgement v
Abstract vi
List of Figures ix
List of Tables x
List of Abbreviations xi
Chapter 1: Introduction **12**
1.1 Introduction and Agricultural Context .. 12
1.2 Aim and Motivation of the Project .. 13
1.3 Scope of the Project 14
1.4 Significant Contributions 14
1.5 Organization of the Report 15
Chapter 2: Literature Review **16**
2.1 Blockchain-Based Supply Chain Traceability .. 16
2.2 AI and Machine Learning for Anomaly Detection .. 17
2.3 IoT-Blockchain Security and Edge Telematics Integration .. 18
2.4 Domain-Specific Agricultural Supply Chain Studies .. 19
2.5 Comparative Analysis of Existing Systems .. 20
2.6 Critical Analysis of Existing Research .. 20
2.7 Identified Research Gaps 21
Chapter 3: Problem Statement and System Requirements .. **21**
3.1 Problem Statement 21
3.2 System Objectives 22
3.3 Functional Requirements 23
3.4 Non-Functional Requirements 24
3.5 Target Stakeholder Profiles and Personas .. 24
Chapter 4: Proposed Methodology and System Design .. **25**
4.1 System Overview and 5-Tier Architecture .. 25
4.2 Operational Flowchart of Proposed Methodology .. 26
4.3 Data Flow Analysis (DFD Level 0 and Level 1) .. 27
4.4 Database Design and Entity-Relationship Modeling .. 29
4.5 Use Case Analysis and Role Governance .. 31
4.6 AI-Based Trust Layer Pipeline 32
4.7 Modular Smart Contracts and Circuit Breakers .. 33
4.8 Dual Persistence and Decentralized Document Pinning .. 34
4.9 Software and System Specifications .. 34
4.10 Dynamic Crop Policy Engine 34
4.11 Project Timeline and Scheduling .. 34
Chapter 5: Results and Discussion **36**

5.1 AI Trust Layer Performance Evaluation .. 36
5.2 Telemetry Fault Detection and Quarantine Isolation .. 37
5.3 Automated Testing and Capstone System Verification .. 37
5.4 Smart-Contract Gas Optimization Benchmarking .. 38
5.5 Multi-Role Platform Dashboards and Traceability .. 38
5.6 Overall Discussion and Research Validation .. 44
Chapter 6: Conclusions and Future Scope .. 46
6.1 Conclusion 46
6.2 Future Scope 46
Appendices 48
Appendix I: System Operational Workflow and State Machine .. 48
Appendix II: AI Trust Layer Verdict Rules and Anomaly Taxonomy .. 49
Bibliography / References 50

List of Figures

Figure 4.1: Four-Tier System Architecture of the AgriChain Platform 25
Figure 4.2: Operational Flowchart of the Proposed Methodology 26
Figure 4.3: Data Flow Diagram (DFD) Level 0: Context Diagram 28
Figure 4.4: Data Flow Diagram (DFD) Level 1: Complete System Data Flow 29
Figure 4.5: Entity-Relationship Diagram (ERD) of AgriChain Relational and On-Chain Schema .. 30
Figure 4.6: Use Case Diagram for AgriChain Multi-Stakeholder Ecosystem 31
Figure 4.7: Project Timeline and Implementation Gantt Chart 35
Figure 5.1: Confusion Matrix of the AI-Based Anomaly Detection Model 36
Figure 5.2: AI-Based Telemetry Anomaly Detection and Quarantine Isolation 37
Figure 5.3: Farmer Dashboard for Agricultural Product and Batch Registration 39
Figure 5.4: Logistics Dashboard for Product Dispatch and Tracking 40
Figure 5.5: Logistics Interface for Product Movement and Custody Updates 40
Figure 5.6: AI Trust Layer for IoT Telemetry Validation and Checkpoints 41
Figure 5.7: Dark-Store Dashboard for Product Receiving and Inventory Management ... 42
Figure 5.8: Administrative Dashboard for Supply Chain Governance and Circuit Breakers .. 42
Figure 5.9: Consumer Interface for QR-Based Product Traceability 43
Figure 5.10: Farm-to-Fork Product Journey with Cumulative Price Breakdown 43
Figure 5.11: Digital Validation Certificate for Product Provenance 44

List of Tables

Table 1.1: List of Abbreviations xi
Table 2.1: Literature Survey: Blockchain-Based Supply Chain Traceability 17
Table 2.2: Literature Survey: AI and Machine Learning for Anomaly Detection 18
Table 2.3: Literature Survey: IoT and Blockchain Security Integration 19
Table 2.4: Literature Survey: Domain-Specific Agricultural Supply Chain Studies ... 20
Table 2.5: Comparative Analysis: Existing Systems vs. Proposed AgriChain System ... 20
Table 4.1: Software and System Specifications 34
Table 4.2: Crop Policy Reference Thresholds for Cold Chain Monitoring 34
Table 5.1: Recorded Performance Metrics of the AI Trust Layer 36
Table 5.2: Recorded End-to-End System Verification Outcomes (PRD Section 6 Suite) .. 38
Table 5.3: Recorded Gas Consumption for Individual vs. Batched Condition Writes ... 38
Table 5.4: Multi-Stage Fair Price Distribution Across Supply Chain Actors 43

List of Abbreviations

Table 1.1: List of Abbreviations

Abbreviation	Full Description / Meaning
AI	Artificial Intelligence
IoT	Internet of Things
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
RBAC	Role-Based Access Control
EVM	Ethereum Virtual Machine
IPFS	InterPlanetary File System
CID	Content Identifier (Cryptographic Hash)
GRN	Goods Receipt Note / Inbound Receiving
ML	Machine Learning
SHA	Secure Hash Algorithm (e.g., SHA-256)
GPS	Global Positioning System
API	Application Programming Interface
REST	Representational State Transfer
UI	User Interface
SCM	Supply Chain Management
XAI	Explainable Artificial Intelligence
SHAP	Shapley Additive exPlanations
MITM	Man-in-the-Middle Attack
IoE	Internet of Everything

Chapter 1: Introduction

1.1 Introduction and Agricultural Context

Agriculture forms the primary livelihood for over 54% of India's workforce and contributes approximately 18% to the national Gross Domestic Product (GDP). Despite its vital importance, the post-harvest supply chain faces severe logistical inefficiencies, high levels of perishability, and multi-tier intermediary markups. According to reports from the Ministry of Food Processing Industries (MoFPI) and the Indian Council of Agricultural Research (ICAR), roughly 30% to 40% of fresh fruits and vegetables spoil before reaching retail markets, resulting in estimated annual post-harvest losses exceeding ₹92,000 crore.

A major cause of this post-harvest loss is the lack of continuous, reliable cold-chain monitoring during transportation and warehouse storage. Perishable commodities—including tomatoes, mangoes, leafy vegetables, and dairy—have narrow temperature tolerances. When temperature control fails during highway transit or at distribution centers, produce spoilage accelerates. Because traditional supply chain logistics lack automated and verifiable tracking, temperature breaches often go unnoticed or are concealed by transporters to avoid commercial penalties. As a result, damaged goods frequently reach retail shelves, leading to food waste and safety risks.

In addition to cold-chain failures, agricultural marketing suffers from opaque middleman markups. Between the farmer and the final consumer, goods pass through several intermediaries, including local aggregators, commission agents (arhtiyas), transport operators, wholesale traders, and retail outlets. Each intermediary adds a margin, yet the purchase prices and operational costs remain hidden from both farmers and consumers. Consequently, smallholder farmers often receive only 20% to 28% of the final retail price paid by consumers, despite shouldering production expenses and weather risks. At the same time, consumers pay premium prices without verifiable confirmation of the harvest date, farming origin, or transit handling conditions.

Blockchain technology offers a viable approach to address these transparency challenges. By recording custody transfers and status updates on a decentralized, append-only distributed ledger, blockchain provides a shared, tamper-resistant record for all participants. Smart contracts further allow deterministic execution of business rules, such as role-gated handovers and automated compliance checks, without relying on a central intermediary.

However, applying blockchain directly to sensor-driven supply chains introduces a significant challenge: the 'Garbage In, Garbage Out' (GIGO) problem. While a blockchain ensures that recorded data cannot be altered after the fact, it cannot verify whether the ingested data was accurate when submitted. In cold-chain logistics, telemetry streams originating from transit sensors may contain sudden spikes, communication dropouts, frozen sensor values, or replay attempts. If unverified telemetry is committed directly to smart contracts, the ledger permanently

stores invalid data, leading to incorrect disputes and unnecessary transaction gas costs. Therefore, an intelligent off-chain filtering layer is necessary to inspect and validate telemetry before it is committed on-chain.

1.2 Aim and Motivation of the Project

The primary aim of this Major Project is to design, implement, and evaluate AgriChain—a decentralized agricultural supply chain traceability platform that provides end-to-end provenance, enforces cold-chain compliance through an AI-based Trust Layer, quarantines anomalous telemetry off-chain, and provides transparent multi-stage price tracking.

The motivation for this project arises from several key technical and socio-economic challenges in agricultural logistics:

- **1.** Addressing the Blockchain GIGO Problem: Developing an off-chain AI Trust Layer that screens and validates simulated environmental telemetry before committing updates to public smart contracts.
- **2.** Gas-Efficient Smart Contract Execution: Minimizing Ethereum Virtual Machine (EVM) transaction costs by bitwise packing multiple telemetry fields into a single 256-bit storage slot and supporting batched updates.
- **3.** Tamper-Evident Off-Chain Quarantine: Storing rejected, breached, or anomalous readings in a cryptographic SHA-256 Quarantine Hash Chain, preserving an auditable log without incurring on-chain transaction gas costs.
- **4.** Fair-Share Price Transparency: Recording the farmer's initial price and each intermediary markup on-chain, enabling retail consumers to verify that the farmer received an equitable share of the purchase price.

1.3 Scope of the Project

The functional and architectural scope of the AgriChain platform covers the post-harvest supply chain journey, incorporating the following components:

- • Stakeholder Portals: Role-tailored web interfaces for Farmers, Logistics Transporters, Dark Store Operators, Retail Consumers, and System Administrators.
- • IoT Telemetry Simulation: A software simulation engine generating realistic ambient temperature (°C), relative humidity (%), simulated GPS route waypoints, and sequential timestamps across transit and storage phases.
- • Pre-Commit AI Trust Layer: An off-chain validation pipeline implemented in FastAPI that uses schema validation, plausibility boundaries, dynamic per-crop thermal thresholds, 11-dimensional feature extraction, and an unsupervised Isolation Forest model.
- • Modular Smart Contracts: Solidity 0.8.28 contracts (ProductRegistry, CustodyTransfer, ColdChainMonitor, PolicyConfig) deployed on Polygon Amoy, utilizing OpenZeppelin AccessControl and Pausable circuit breakers.
- • Dual Persistence Architecture: PostgreSQL (Supabase IPv4 Pooler) for fast indexed querying, local SQLite for transaction staging, Pinata IPFS for pinning quality inspection certificates, and an append-only SHA-256 quarantine hash-chain.

- • Presentation Layer: A responsive React 18 single-page application built with Vite and Tailwind CSS, featuring interactive dashboards, dynamic freshness ratings, QR code scanning, and price journey visualizers.

1.4 Significant Contributions

This project presents several practical engineering contributions to decentralized agricultural supply chain architectures:

- **1.** Pre-Commit AI Trust Validation Pipeline: Developed a 7-stage off-chain screening pipeline that achieved a 97.22% overall F1-score and 100% recall across simulated sensor fault scenarios, preventing invalid data from reaching the ledger.
- **2.** Single-Slot Packed EVM Storage: Engineered a bitwise packed Solidity struct for environmental telemetry that fits into a single 256-bit storage word, achieving a 52.71% gas reduction under batched updates (N=20).
- **3.** Cryptographically Auditable Quarantine Hash Chain: Built an off-chain SHA-256 chained quarantine ledger that maintains an immutable audit trail of rejected readings, thermal breaches, and replay attempts without gas overhead.
- **4.** Dynamic Crop Policy Engine: Implemented configurable thermal envelopes for diverse crop varieties (e.g., tomato, mango, wheat), replacing rigid global thresholds with per-crop compliance monitoring.
- **5.** Transparent Cumulative Price Trail: Established an on-chain accounting mechanism that records each actor's price margin during custody handovers, allowing consumers to verify a 50.0% farmer revenue share.
- **6.** Role-Based Access Control and Circuit Breakers: Integrated OpenZeppelin AccessControl for role separation and emergency pause mechanisms to halt state transitions during detected irregularities.
- **7.** Comprehensive End-to-End System Verification: Conducted automated testing across 10 multi-actor integration test cases and 1,000 simulated telemetry cycles, validating system reliability and data flow consistency.

1.5 Organization of the Report

The remainder of this report is organized as follows:

- • Chapter 2 (Literature Review): Surveys existing research in blockchain traceability, AI/ML anomaly detection, edge telematics security, and agricultural supply chain architectures, identifying key research gaps.
- • Chapter 3 (Problem Statement and System Requirements): Formulates the technical problem statement, system objectives, and functional and non-functional requirements.
- • Chapter 4 (Proposed Methodology and System Design): Explains the 5-tier architecture, operational flowchart, DFD Level 0/1 diagrams, database schema, AI trust pipeline, modular smart contracts, and project schedule.
- • Chapter 5 (Results and Discussion): Presents experimental evaluation results, including AI confusion matrices, quarantine dashboards, gas consumption benchmarks, and multi-role dashboard implementations.

- • Chapter 6 (Conclusions and Future Scope): Concludes the report with a summary of achievements and outlines potential future enhancements.
- • Appendices and References: Provides the operational state transition matrix, anomaly decision rules, and cited bibliography.

Chapter 2: Literature Review

The combination of distributed ledger technologies, sensor-based monitoring, and machine learning anomaly detection has received increasing attention as a means to improve supply chain transparency and food safety. To establish the context for the AgriChain platform, this chapter reviews relevant published literature across IEEE, Elsevier, and Springer venues. The review covers four main themes: blockchain-based traceability, AI/ML anomaly detection, edge telematics security, and domain-specific agricultural studies, followed by a comparative analysis and research gaps.

2.1 Blockchain-Based Supply Chain Traceability

Using distributed ledgers in supply chains aims to replace fragmented databases with a shared, tamper-resistant record of custody and product status. Agrawal et al. (2025) [1] explored the integration of the Internet of Everything (IoE) and Artificial Intelligence in supply chains, noting that decentralized tracking reduces information asymmetry between participants. However, they observed that transaction throughput and on-chain storage costs on public blockchains pose scalability challenges for high-frequency data recording.

Maurya et al. (2025) [3] examined blockchain security mechanisms for IoT networks, surveying consensus protocols and access control schemes. Their study highlighted that while Proof-of-Stake (PoS) and Layer-2 networks lower transaction energy and cost, smart contract storage bloat remains a key challenge. Storing frequent environmental readings directly on-chain requires careful data filtering and efficient storage packing.

Dudczyk et al. (2024) [4] conducted a survey of blockchain applications in global supply chain management, finding that while traceability improves stakeholder trust, most deployments lack automated, real-time quality verification mechanisms. Table 2.1 summarizes key literature on blockchain-based traceability architectures.

Table 2.1: Literature Survey: Blockchain-Based Supply Chain Traceability

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications	Dudczyk, Dunston & Crosby (2024)	Blockchain in Global Supply Chains	Reviews blockchain applications and benefits for global supply-chain transparency and traceability.	Lacks agriculture-specific evaluation.
2	Revolutionizing Agri-Food Supply Chain	Aggarwal, Rani, Rani & Sharma (2024)	Blockchain in Agri-Food	Provides blockchain-based	Scalability concerns in large-scale

	Management with Blockchain-Based Traceability and Navigation Integration			traceability and navigation for food provenance and consumer trust.	deployment.
3	High-Efficiency Blockchain-Based Supply Chain Traceability	Wu, Jiang & Cao (2023)	Blockchain Traceability	Improves efficiency of blockchain-based supply-chain traceability by reducing search latency and storage overhead.	Limited real-world/large-scale validation.
4	Integrating Blockchain Technology for Enhanced Transparency and Security in Supply Chains	Chaker & Damak (2024)	Blockchain for Supply Chain	Discusses blockchain integration for supply-chain transparency and security.	Mostly conceptual; limited quantitative evidence.
5	Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Blockchain in Organic Food	Examines how blockchain can improve trust in organic food supply-chain relationships.	Findings are context-specific to organic food.

2.2 AI and Machine Learning for Anomaly Detection

Machine learning models for anomaly detection are essential to screen out faulty or malicious readings from sensor data streams. Mohamed et al. (2025) [2] presented an Adaptive WavePCA-Autoencoder (AWPA) combined with an Adaptive Hybrid Exploit Detection Network (AHEDNet) for identifying security anomalies in sensor streams. Although the deep learning model showed high accuracy, its computational and memory requirements make it less suitable for low-latency API microservices running on standard servers.

Zhou et al. (2024) [13] evaluated Long Short-Term Memory (LSTM) networks and Support Vector Machines (SVM) for detecting data tampering in supply chain workflows. While recurrent models capture sequential relationships well, they require extensive labeled training datasets and are susceptible to concept drift. In contrast, unsupervised methods such as Isolation Forest isolate anomalies by recursively partitioning feature values. Because anomalous observations require

fewer random partitions to isolate than normal points, Isolation Forest operates with linear time complexity $O(n)$ and low memory overhead, making it practical for real-time telemetry validation.

Kandasamy and Roseline (2025) [14] investigated hybrid deep learning frameworks for detecting data injection attacks in sensor networks, demonstrating that multi-dimensional feature engineering improves sensitivity to gradual sensor drift. Table 2.2 summarizes AI and ML approaches to anomaly detection.

Table 2.2: Literature Survey: AI and Machine Learning for Anomaly Detection

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Optimizing Supply Chain Management With IoE and AI	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	IoE & AI in SCM	Explores AI and Internet of Everything for supply-chain optimization and decision-making.	Lacks empirical validation in agricultural contexts.
2	Zero-Day Exploits Detection with Adaptive WavePCA-Autoencoder (AWPA) Adaptive Hybrid Exploit Detection Network (AHEDNet)	Mohamed, Al-Saleh, Sharma & Tejani (2025)	ML-Based Security	Uses a hybrid ML approach for proactive zero-day exploit detection.	Requires high-quality, large training datasets.
3	A Novel Malware Detection Model in the Software Supply Chain Based on LSTM and SVMs	Zhou, Li, Fu & Jiao (2024)	ML Malware Detection	Uses LSTM and SVM-based models for malware detection.	Higher model complexity increases deployment cost.
4	Harnessing Advanced Hybrid Deep Learning Model for Real-Time Detection and Prevention of Man-in-the-Middle Cyber Attacks	Kandasamy & Roseline (2025)	Deep Learning Security	Applies hybrid deep learning for real-time MITM attack detection and prevention.	Computationally intensive for constrained devices.

2.3 IoT-Blockchain Security and Edge Telematics Integration

The interface between edge telematics modules and decentralized networks is a critical consideration in supply chain architectures. Yavaprabhas et al. (2024) [5] conducted a field study on stakeholder trust in organic food supply chains using blockchain. Their findings indicated that despite the presence of smart contracts, consumers and auditors remained cautious regarding manual data inputs and unvalidated sensor feeds, emphasizing the need for automated pre-commit screening.

Aggarwal et al. (2024) [6] integrated GPS corridor tracking with smart contracts to monitor perishable agri-food shipments, demonstrating route compliance tracking but noting challenges in handling transmission dead-zones. Mangala et al. (2024) [9] and Eghmazi et al. (2024) [12] explored decentralized document pinning and hash chaining for IoT security, showing that storing cryptographic Content Identifiers (CIDs) on-chain while keeping large files on IPFS provides verifiable tamper evidence at low cost. Table 2.3 synthesizes research concerning IoT-blockchain security integration.

Table 2.3: Literature Survey: IoT and Blockchain Security Integration

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Secure Pharmaceutical Supply Chain Using Blockchain in IoT Cloud Systems	Mangala, Naveen, Reddy, Buyya, Venugopal, Iyengar & Patnaik (2024)	Blockchain + IoT	Integrates blockchain with IoT-cloud systems for secure authentication and anti-counterfeiting.	High system complexity limits adoption.
2	Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT-Enabled SCM	Shows how IoT improves supply-chain resilience, efficiency and coordination.	Data privacy risks remain unresolved.
3	Supply Chain Management Growth With the Adoption of Blockchain Technology (BoT) and Internet of Things (IoT)	Singh & Dadhich (2023)	Blockchain + IoT	Presents a model showing improved supply-chain coordination and visibility through Blockchain + IoT.	Lacks real implementation and empirical testing.
4	Blockchain-	Maurya,	IoT Security	Reviews	Introduces

	Driven Security for IoT Networks: State-of-the-Art, Challenges and Future Directions	Rishiwal, Yadav, Shiblee, Yadav, Agarwal & Chaudhry (2025)		blockchain-based approaches for improving IoT security and data protection.	computational overhead.
5	Enhancing IoT Data Security: Using the Blockchain to Boost Data Integrity and Privacy	Eghmazi, Ataei, Landry & Chevrette (2024)	IoT Data Security	Uses blockchain to improve IoT data integrity, privacy and security.	Scalability with high sensor-data volume is not fully addressed.

2.4 Domain-Specific Agricultural Supply Chain Studies

Agricultural goods present distinct challenges due to perishability, seasonal harvesting, and biological decay kinetics. Wu, Jiang, and Cao (2023) [7] proposed a high-efficiency blockchain traceability framework with optimized state verification, showing that data encoding optimizations reduce verification latency. However, their study applied uniform threshold limits, whereas optimal storage conditions differ substantially across crops (e.g., tomatoes require +2°C to +8°C, while tropical mangoes require +10°C to +15°C).

Chaker and Damak (2024) [8] examined transparency in agri-food supply chains, observing that intermediaries often resist traceability systems unless price transparency mechanisms are clearly defined. Shoomal et al. (2024) [10] and Singh and Dadhich (2023) [11] emphasized that supply chain resilience requires combining environmental monitoring with contract-enforced custody transitions. Table 2.4 outlines domain-specific agricultural studies.

Table 2.4: Literature Survey: Domain-Specific Agricultural Supply Chain Studies

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration	Aggarwal, Rani, Rani & Sharma (2024)	Agri-Food Traceability	Improves end-to-end food provenance, traceability and consumer trust.	Scalability concerns in large-scale deployment.
2	Demystifying the Impact of Blockchain on	Yavaprabhas, Kurnia, Seyedghorban	Organic Food Supply Chain	Demonstrates blockchain's role in	Findings are specific to organic food

	Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains	& Samson (2024)		strengthening trust within organic food supply chains.	case studies.
3	Optimizing Supply Chain Management With IoE and AI	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	AI in Supply Chain	Explores AI and IoE for supply-chain optimization.	Lacks empirical validation in agricultural contexts.
4	Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT Supply Chain	Highlights IoT applications for real-time tracking and improved supply-chain resilience.	Privacy, interoperability and scalability challenges.
5	Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications	Dudczyk, Dunston & Crosby (2024)	Global Supply Chain	Identifies blockchain's benefits for transparency and traceability across supply chains.	Does not specifically evaluate agricultural applications.

2.5 Comparative Analysis of Existing Systems

To highlight the specific features of AgriChain, Table 2.5 presents a comparison between existing supply chain implementations and our proposed 5-tier architecture across key parameters: system architecture, sensor validation, contract modularity, price transparency, off-chain storage integrity, and corrupted telemetry handling.

Table 2.5: Comparative Analysis: Existing Systems vs. Proposed AgriChain System

Dimension / Parameter	Existing Supply Chain Systems	Proposed AgriChain Framework
System Architecture	Mainly focus on blockchain-based traceability in isolation	Unified 5-tier architecture: Blockchain + Simulated IoT Telematics + AI Trust Layer + Smart Contracts + React UI
Sensor Data Validation	IoT data directly recorded on-chain without validation ('Garbage In,	Multi-stage AI Trust Layer validates telemetry streams off-chain before

	Garbage Out')	committing to blockchain
Smart Contract Customization	Monolithic, rigid contracts with limited parameter flexibility	Modular smart contracts (ProductRegistry, CustodyTransfer, ColdChainMonitor, PolicyConfig) with OpenZeppelin RBAC and Pausable circuit breakers
Payment & Price Transparency	Opaque intermediary markups; payments unlinked to verified custody	Transparent, cumulative on-chain price trail recording farmer gate price and actor margins at every handoff
Off-Chain Storage & Data Integrity	Unprotected off-chain databases vulnerable to silent tampering	Tamper-evident SHA-256 Quarantine Hash-Chain and decentralized IPFS document vault with on-chain CIDs
Handling of Corrupted Telemetry	Thermal breaches or sensor faults pollute public ledger	Anomalies, sensor freezes, and replay attacks isolated into cryptographic quarantine without wasting gas

2.6 Critical Analysis of Existing Research

A critical review of the literature reveals five recurring limitations in current blockchain supply chain implementations:

- **1.** Lack of Pre-Commit Telemetry Screening: Many existing systems stream sensor data directly to smart contracts without off-chain validation, leading to ledger pollution when sensors experience temporary spikes or transmission errors.
- **2.** High On-Chain Storage Costs: Storing separate 256-bit EVM words for each environmental metric (temperature, humidity, coordinates, timestamp) leads to high gas consumption, making frequent updates impractical.
- **3.** Monolithic Contract Structures: Relying on a single large contract for registration, tracking, policy enforcement, and access control complicates upgrades and increases security risk.
- **4.** Unverifiable Off-Chain Logging: Systems that discard anomalous readings into standard relational databases lack cryptographic tamper evidence, leaving logs vulnerable to alteration.
- **5.** Lack of Intermediary Price Accountability: While many platforms track geographical origin, few record intermediate prices at each custody handover, leaving middleman markups untracked.

2.7 Identified Research Gaps

Based on these limitations, the following research gaps (RG) were identified and addressed in this project:

- • RG-1: Need for a lightweight, pre-commit AI validation pipeline capable of filtering high-frequency telemetry under sub-second latency constraints without requiring dedicated GPU infrastructure.
- • RG-2: Need for an EVM storage packing strategy that fits multiple environmental metrics into a single 256-bit slot to reduce condition logging gas costs.

- • RG-3: Need for an append-only, cryptographic hash-chain ledger dedicated to logging quarantined readings off-chain with tamper evidence.
- • RG-4: Need for configurable, per-crop cold-chain policy management that supports cultivar-specific environmental bounds.
- • RG-5: Need to integrate decentralized document pinning (IPFS) with on-chain role-based access control and emergency pause circuit breakers.
- • RG-6: Need for an on-chain cumulative price trail that tracks intermediate actor markups and allows consumers to verify fair producer compensation.
- • RG-7: Need for end-to-end verification across simulated sensor anomalies, network latency conditions, and multi-actor supply chain workflows.

Chapter 3: Problem Statement and System Requirements

3.1 Problem Statement

Agricultural supply chains often suffer from post-harvest spoilage, lack of verifiable cold-chain surveillance, and opaque intermediary pricing that disadvantages smallholder farmers. While distributed ledgers provide tamper-resistant history, ingesting raw telemetry directly into smart contracts leads to the 'Garbage In, Garbage Out' (GIGO) problem: corrupted, drifted, or replayed readings are permanently committed to the ledger, causing false disputes and wasting transaction gas. Additionally, monolithic contract structures result in high gas costs, and off-chain anomaly logs typically lack cryptographic tamper evidence.

Formally, let an agricultural produce batch B_i be registered at time t_0 by farmer A_{farmer}. As B_i transitions through sequential custody states $S = \{S_{\text{REG}}, S_{\text{TRANSIT}}, S_{\text{STORAGE}}, S_{\text{RETAIL}}, S_{\text{SOLD}}\}$, a simulated edge telematics client generates a time-series telemetry sequence $T_i = \{m_1, m_2, \dots, m_k\}$, where each measurement tuple is defined as $m_k = (\tau_k, \theta_k, \phi_k, \lambda_k, \mu_k)$ representing timestamp τ_k, temperature θ_k, humidity ϕ_k, latitude λ_k, and longitude μ_k.

The technical objective is to design an automated verification operator $\mathcal{V}(m_k \mid \mathcal{P}_c)$ conditioned on dynamic crop policy $\mathcal{P}_c = (\theta_{\min}, \theta_{\max}, \phi_{\min}, \phi_{\max})$ such that:

- **1.** Telemetry Validation: If m_k satisfies plausibility limits and yields an Isolation Forest anomaly score $s(m_k) \le \gamma_{\text{threshold}}$, it is batched and committed to smart contract state $C_{\text{ColdChain}}$ on Polygon Amoy in a single packed 256-bit slot.
- **2.** Quarantine Routing: If m_k violates crop bounds or triggers $s(m_k) > \gamma_{\text{threshold}}$, m_k is quarantined in an off-chain SHA-256 hash-chain $H_k = \text{SHA-256}(H_{k-1} \parallel m_k \parallel \text{Reason}_k \parallel \tau_k)$ without public ledger contamination.
- **3.** Custody and Price Progression: Custody handovers $S_k \to S_{k+1}$ are strictly forward-only and governed by AccessControl, atomically updating an immutable price vector $\mathbf{P}_i = (p_{\text{farmer}}, p_{\text{logistics}}, p_{\text{store}}, p_{\text{retail}})$ to provide transparent margin visibility.

3.2 System Objectives

To address these requirements, the AgriChain platform was developed to achieve six specific engineering objectives:

- • Objective 1: Multi-Stage AI Trust Engine: Develop a lightweight pre-commit validation pipeline capable of processing simulated telemetry in under 150 milliseconds with an anomaly detection F1-score exceeding 95%.
- • Objective 2: Gas-Optimized Smart Contracts: Design modular EVM smart contracts utilizing bitwise packed storage to achieve a gas reduction of over 45% compared to multi-word storage implementations.
- • Objective 3: Cryptographic Quarantine Ledger: Build an off-chain SHA-256 hash-chain audit log that captures all rejected or anomalous readings, ensuring auditability without on-chain gas costs.
- • Objective 4: Dynamic Crop-Specific Governance: Implement configurable on-chain and off-chain policy thresholds supporting distinct thermal envelopes for different crop varieties.
- • Objective 5: Transparent Fair Price Accounting: Engineer an on-chain cumulative price trail that records intermediate actor handoffs and verifies an equitable farmer revenue share.
- • Objective 6: Responsive Multi-Role Web Interface: Build dedicated React 18 dashboards for Farmers, Logistics Transporters, Dark Store Operators, Consumers, and Admins, supporting real-time visualizers and QR code verification.

3.3 Functional Requirements

The system specifications are organized into twelve functional requirements (FR):

- • FR-1 (Actor Authentication & RBAC): Enforce role-based access control mapping Ethereum addresses and backend JWT sessions to discrete roles (Farmer, Logistics, DarkStore, Consumer, Admin).
- • FR-2 (Batch Produce Registration): Allow farmers to register harvest batches with crop variety, harvest date, quantity in kg, base price in integer paise, and farm location.
- • FR-3 (Automated QR Code Generation): Dynamically generate QR codes encoding unique batch identifiers and direct provenance verification URLs upon batch creation.
- • FR-4 (Simulated Telemetry Ingestion): Accept simulated JSON telemetry streams and execute schema checks, physical bounds bounding, and sequential timestamp verification.
- • FR-5 (AI Anomaly Scoring): Extract 11 statistical features from telemetry and evaluate an Isolation Forest model to assign an anomaly score and validation status.
- • FR-6 (Quarantine Logging): Record anomalous or out-of-policy readings into an append-only SHA-256 hash-chain, linking each entry to the previous block digest.
- • FR-7 (Packed On-Chain Relay): Relay validated readings to the ColdChainMonitor smart contract using packed struct parameters, emitting TelemetryLogged events.
- • FR-8 (Forward-Only Custody Progression): Enforce a strictly forward-only custody sequence (REGISTERED → IN_TRANSIT → IN_STORAGE → AT_RETAIL → SOLD), preventing out-of-order handovers.
- • FR-9 (Multi-Tier Price Logging): Atomically record each recipient's price addition in integer paise during custody handovers, maintaining the cumulative retail price.

- • FR-10 (IPFS Document Pinning): Pin quality inspection reports and organic certificates to IPFS via Pinata, anchoring resulting CIDs on-chain.
- • FR-11 (Emergency Circuit Breakers): Provide administrative controls to pause contract state transitions during detected irregularities via Pausable circuit breakers.
- • FR-12 (Consumer Provenance Portal): Allow consumers to scan product QR codes to view the farm-to-fork journey, dynamic freshness grade, fair price breakdown, and verified reviews.

3.4 Non-Functional Requirements

The platform architecture satisfies eight non-functional requirements (NFR):

- • NFR-1 (Validation Latency): The AI Trust Layer pre-commit validation must execute feature extraction and anomaly scoring in under 150 ms per telemetry payload.
- • NFR-2 (EVM Gas Efficiency): Batched condition updates (N=20) must average less than 50,000 gas per reading, achieving at least 45% savings over uncompressed storage.
- • NFR-3 (Tamper Evidence): Any retroactive alteration to quarantined records must be mathematically detectable via hash-chain verification sweeps within milliseconds.
- • NFR-4 (Database Performance): Relational storage must support concurrent read/write operations using connection pooling and SQLite WAL mode.
- • NFR-5 (Responsive UI Layout): The web interface must maintain a responsive layout across mobile, tablet, and desktop viewports (320px to 2560px).
- • NFR-6 (Security Standards): Passwords must be hashed using bcrypt, API sessions managed via HMAC-SHA256 JWTs, and smart contracts aligned with audited OpenZeppelin v5.0 libraries.
- • NFR-7 (Asynchronous Concurrency): The backend must utilize non-blocking asynchronous I/O across database access, IPFS uploads, and blockchain transactions.
- • NFR-8 (Integer Precision Rules): All financial figures must use integer paise (1 INR = 100 paise) and temperatures must use integer deci-degrees Celsius (24.5°C = 245) to avoid floating-point rounding issues.

3.5 Target Stakeholder Profiles and Personas

The platform interfaces were designed to accommodate five distinct user personas representing typical supply chain participants:

- • Persona 1: Ramesh Patel (Smallholder Farmer): Cultivates a 4-acre organic farm near Nashik. Needs an easy-to-use crop registration form, proof of harvest origin, and transparent price tracking without technical complexity.
- • Persona 2: Vikram Singh (Refrigerated Fleet Transporter): Operates refrigerated transport vehicles. Needs continuous telemetry tracking, route milestone logging, and unambiguous custody handovers to protect against wrongful damage claims.
- • Persona 3: Priya Sharma (Dark Store Hub Manager): Manages an urban distribution hub. Requires incoming shipment verification, storage bay allocation, and prompt inventory processing for local fulfillment.

- • Persona 4: Ananya Iyer (Retail Consumer): Urban grocery buyer seeking verifiable information on harvest date, chemical residue testing, transit handling, and fair farmer compensation.
- • Persona 5: Rajesh Verma (Supply Chain Administrator): System administrator overseeing contract roles, monitoring quarantine alerts, auditing hash-chain integrity, and activating circuit breakers if necessary.

Chapter 4: Proposed Methodology and System Design

4.1 System Overview and 5-Tier Architecture

The proposed AgriChain platform is organized as a modular 5-tier architecture that separates data simulation and capture, pre-commit AI validation, smart contract execution on blockchain, off-chain storage, and the web-based presentation layer. This decoupled structure provides clean separation of concerns, failure isolation between layers, and maintainable software components. Figure 4.1 illustrates the overall 5-tier architecture of the platform.

The 5-tier architecture coordinates data flow and lifecycle operations across the following layers:

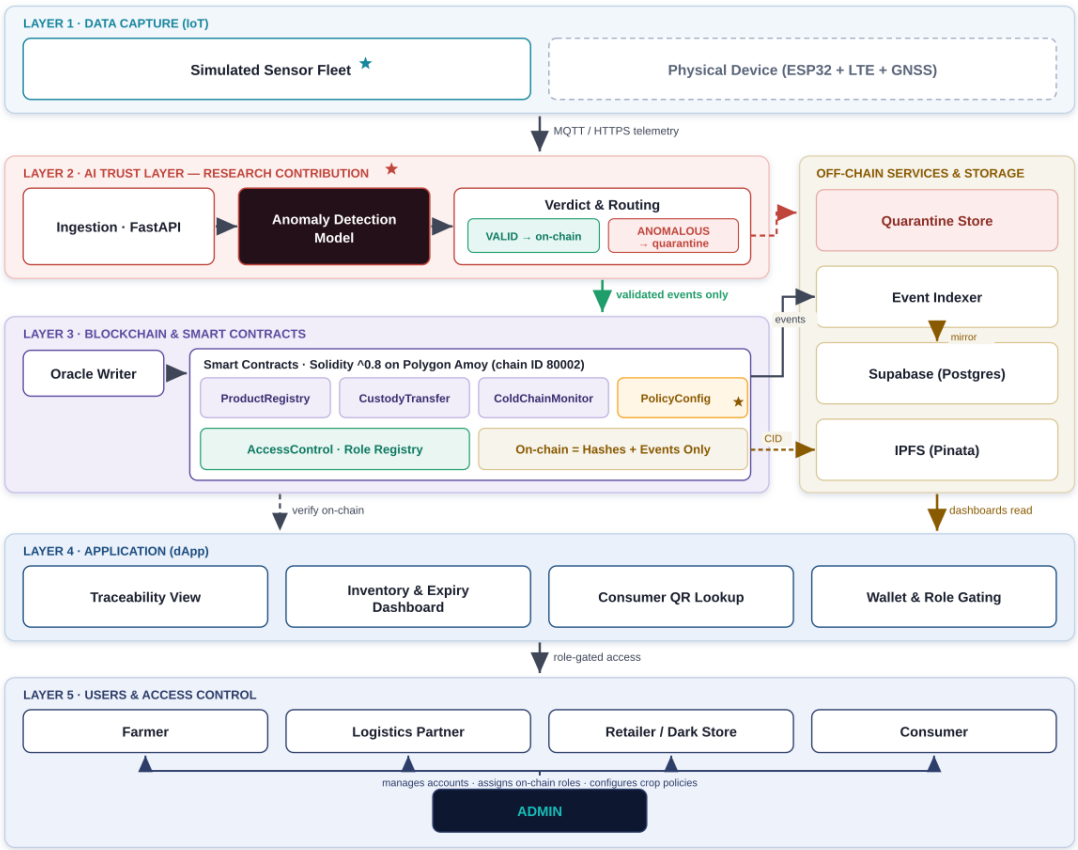


Figure 4.1: Four-Tier System Architecture of the AgriChain Platform

- • Tier 1: IoT Telemetry & Edge Simulation Layer: Generates high-frequency environmental data via simulated IoT edge nodes and a high-fidelity Python telemetry simulation engine. Monitored parameters include simulated ambient temperature (°C), relative humidity (%), synthetic GPS geographical coordinates (latitude and longitude), device identifiers, and sequential timestamps. Telemetry packets are serialized as encrypted JSON payloads and transmitted over HTTPS/REST to the backend ingestion gateway.
- • Tier 2: AI Trust Layer (FastAPI Ingestion Engine): Serves as a pre-commit gatekeeper executing off-chain validation prior to blockchain submission. The engine executes a 7-stage verification pipeline comprising Pydantic v2 schema validation,

physical and temporal plausibility bounding, dynamic crop policy threshold compliance, 11-dimensional feature extraction, unsupervised Isolation Forest anomaly scoring, and SHA-256 event fingerprinting. Readings exhibiting sensor freezes, thermal breaches, or replay signatures are isolated into a cryptographic quarantine ledger without consuming on-chain gas.

- • Tier 3: Blockchain & Smart Contract Layer (Polygon Amoy / Hardhat): Manages decentralized state transitions and enforces business logic through four modular Solidity 0.8.28 contracts (ProductRegistry, CustodyTransfer, ColdChainMonitor, PolicyConfig). Utilizing OpenZeppelin v5.0 libraries, this tier provides Role-Based Access Control (RBAC) and Pausable emergency circuit breakers. Environmental conditions are persisted using a bitwise single-slot packed struct, achieving substantial gas reductions.
- • Tier 4: Off-Chain Storage & Decentralized Assets Layer: Uses a dual-engine database setup: PostgreSQL (Supabase IPv4 Pooler) for fast indexed querying and SQLite with Write-Ahead Logging (WAL) for local transaction staging. Quality inspection reports and organic certificates are pinned to IPFS via the Pinata Gateway, returning immutable Content Identifiers (CIDs) anchored on-chain. This tier also hosts the append-only SHA-256 Quarantine Hash Chain.
- • Tier 5: Multi-Role Presentation Layer: A responsive React 18 single-page application built with Vite and Tailwind CSS. It provides role-tailored interfaces for Farmers, Logistics Transporters, Dark Store Operators, and System Administrators, alongside a publicly accessible mobile-responsive Consumer Provenance Portal supporting QR code batch verification.

4.2 Operational Flowchart of Proposed Methodology

The operational lifecycle of produce batches follows a deterministic, forward-only progression, ensuring that product movement, telemetry surveillance, AI verification, and financial handoffs remain synchronized across all actors. Figure 4.2 presents the operational flowchart from harvest registration to consumer retail purchase.

The operational workflow unfolds across seven sequential phases:

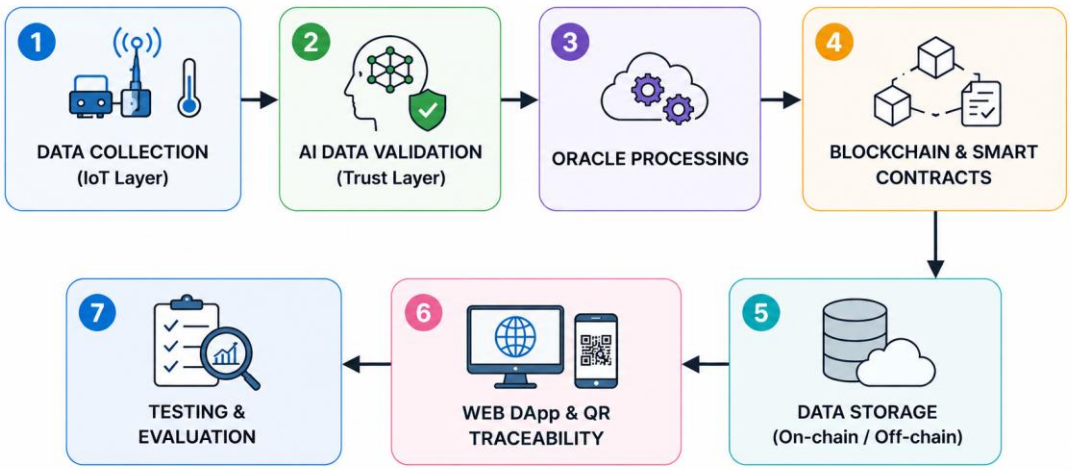


Figure 4.2: Operational Flowchart of the Proposed Methodology

- **1.** Phase 1: Produce Registration: The farmer logs in to the Farmer Portal, entering crop cultivar, harvest quantity (kg), farm location, and base gate price in integer paise. The ProductRegistry contract mints an on-chain batch record, and the frontend dynamically generates a batch QR code.
- **2.** Phase 2: IoT Telemetry Simulation Initiation: When the batch moves to transit status, a simulated edge telematics client is paired with the batch ID, streaming realistic environmental telemetry packets at regular intervals (e.g., every 30 seconds) along a simulated transit corridor.
- **3.** Phase 3: Pre-Commit AI Auditing: The FastAPI AI Trust Layer intercepts incoming telemetry packets and executes the 7-stage validation pipeline, evaluating statistical plausibility, crop thermal limits, and Isolation Forest anomaly scores.
- **4.** Phase 4: Cryptographic Routing & Persistence: Validated telemetry is written to relational storage and queued for packed on-chain condition logging. Anomalous or out-of-policy readings are appended to the SHA-256 Quarantine Hash Chain with explainable reason codes.
- **5.** Phase 5: Custody Transfer & Price Accumulation: Upon arrival at the dark-store distribution hub, the logistics driver initiates a custody transfer. The dark-store manager verifies the batch, enters the handling fee, and confirms acceptance, advancing the batch state from IN_TRANSIT to IN_STORAGE.
- **6.** Phase 6: Quality Inspection & IPFS Pinning: Quality assurance personnel conduct grading and upload certified laboratory inspection documents to IPFS, anchoring the resulting CID to the batch record on-chain.
- **7.** Phase 7: Consumer Provenance Verification: At the point of retail purchase, the consumer scans the product QR code to inspect the complete farm-to-fork journey, dynamic freshness grade, verified price breakdown, and IPFS laboratory certificates.

4.3 Data Flow Analysis

Data flow diagrams (DFD) provide a structured representation of information transit, process transformations, and data stores governing the AgriChain platform across Context Level 0 and Detailed System Level 1.

4.3.1 DFD Level 0: Context Diagram

The Context Diagram (Level 0) defines the overall system boundary, illustrating the external entities that interact with the core AgriChain processing engine, along with the primary data inflows and outflows. Figure 4.3 illustrates the DFD Level 0 Context Diagram.

The Context Diagram identifies four primary external entities interacting with the platform:

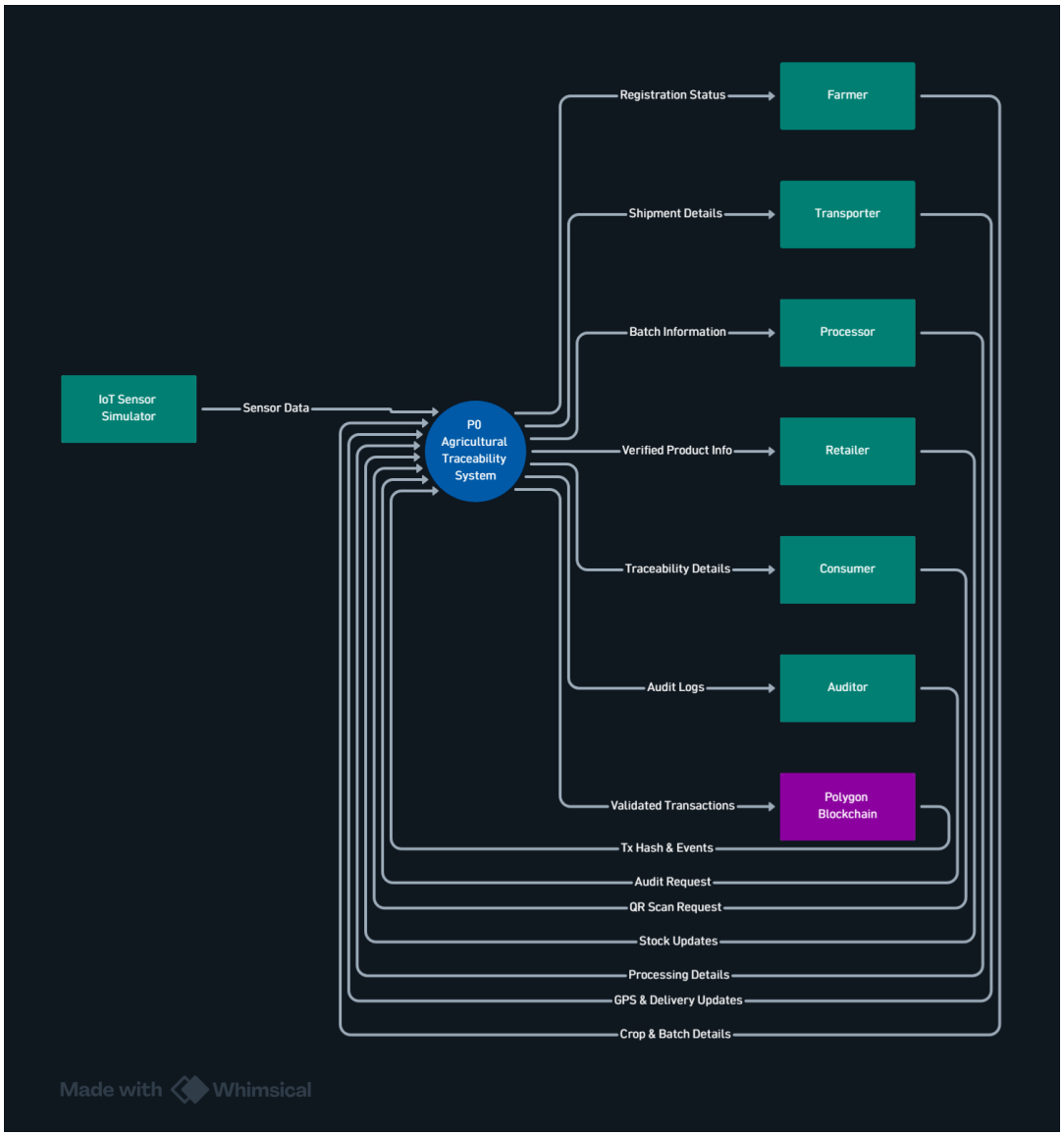


Figure 4.3: Data Flow Diagram (DFD) Level 0: Context Diagram

- • E1 (Simulated IoT Edge Nodes / Telematics Simulator): Transmits simulated environmental telemetry payloads comprising temperature, humidity, GPS coordinates, and sequence timestamps to the central ingestion gateway.
- • E2 (Supply Chain Actors): Encompasses Farmers, Logistics Transporters, and Dark Store Operators who submit authentication credentials, batch registration forms, custody handover confirmations, and margin updates.
- • E3 (Consumers & Regulators): Queries batch provenance via mobile QR scans, retrieving journey timelines, freshness ratings, fair price trails, and inspection certificates.
- • E4 (Polygon Amoy Blockchain & IPFS): Receives role-gated smart contract transactions, packed condition records, and document CIDs, returning transaction receipts and event logs.

4.3.2 DFD Level 1: Complete System Data Flow

The Level 1 Data Flow Diagram decomposes the system into five primary subprocesses and five data repositories, showing the routing of verified versus quarantined information. Figure 4.4 presents the complete Level 1 DFD.

The Level 1 diagram details the flow of data across five primary functional processes:

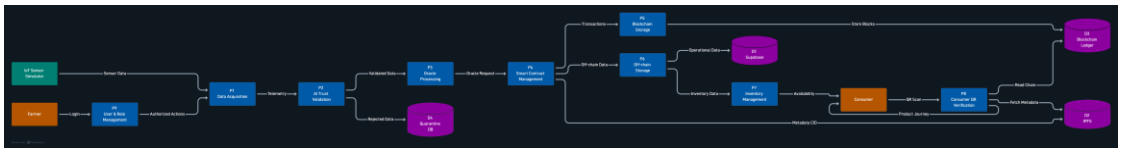


Figure 4.4: Data Flow Diagram (DFD) Level 1: Complete System Data Flow

- • Process 1.0 (Telemetry Ingestion & AI Trust Validation): Ingests simulated sensor streams from E1, queries dynamic crop thresholds from Data Store D5 (Policy Cache), extracts statistical features, executes Isolation Forest evaluation, and directs data into verified streams or quarantine records.
- • Process 2.0 (Smart Contract & Blockchain Relaying): Packs verified telemetry into single-slot bitwise representations, signs transactions via backend relayer accounts, and commits updates to Data Store D3 (Polygon Blockchain).
- • Process 3.0 (Off-Chain Storage & Quarantine Engine): Writes raw readings and state events to Data Store D1 (PostgreSQL/SQLite Relational DB) while cryptographically linking anomalous readings into Data Store D2 (Quarantine Hash Chain).
- • Process 4.0 (Multi-Actor Dashboard & Access Control): Authenticates supply chain actors (E2), validates RBAC roles, coordinates custody transfers, and pins uploaded inspection documents to Data Store D4 (IPFS Document Pins).
- • Process 5.0 (Consumer Provenance Verification): Resolves batch queries from consumers (E3), fetches on-chain custody states and price trails from D3, retrieves journey telemetry from D1, verifies IPFS certificates from D4, and computes dynamic freshness grades.

4.4 Database Design and Entity-Relationship Modeling

The platform uses a normalized relational database schema designed to mirror on-chain smart contract state while providing fast querying for frontend dashboards and geographical visualization. Figure 4.5 illustrates the Entity-Relationship Diagram (ERD).

The database architecture is structured across seven core relational entities:



Figure 4.5: Entity-Relationship Diagram (ERD) of AgriChain Relational and On-Chain Schema

- • USER / ACTOR (user_id PK, wallet_address, role_enum, full_name, email, organization_name, created_at): Stores stakeholder credentials, cryptographic wallet addresses, and assigned RBAC permissions.
- • BATCH (batch_id PK, crop_name, quantity_kg, farm_location, base_price_paise, current_holder_id FK, current_status_enum, contract_batch_id, created_at): Maintains produce batch identity and lifecycle status.
- • CUSTODY_EVENT (event_id PK, batch_id FK, from_holder_id FK, to_holder_id FK, event_type, transaction_hash, actor_price_paise, cumulative_price_paise, timestamp): Logs each custody transition and price increment.
- • READING (reading_id PK, batch_id FK, temperature_deci_c, humidity_pct, latitude, longitude, trust_score, is_anomalous, validation_verdict, timestamp): Persists simulated environmental telemetry records.
- • QUARANTINE_LOG (log_id PK, reading_id FK, reason_code, anomaly_score, previous_hash, current_hash, is_verified, timestamp): Houses cryptographically chained rejected sensor data for auditability.
- • POLICY (policy_id PK, crop_name, min_temp_deci_c, max_temp_deci_c, min_humidity_pct, max_humidity_pct, max_excursion_minutes): Encapsulates cultivar-specific environmental bounds governing compliance.
- • BATCH_DOCUMENT (doc_id PK, batch_id FK, doc_type, ipfs_cid, sha256_hash, upload_timestamp): Tracks laboratory quality certificates pinned to IPFS.

4.4.1 Tamper-Evident SHA-256 Quarantine Hash-Chain Algorithm

To ensure that rejected telemetry cannot be retroactively modified or deleted, the quarantine repository is implemented as an append-only SHA-256 cryptographic

hash-chain. The genesis block is initialized with a 32-byte zero-digest: $H_0 = \text{0x000...0}$. For each subsequent quarantined reading $k \ge 1$, the digest H_k is computed recursively as follows:

$$H_k = \text{SHA-256}\Big(H_{k-1} \parallel \text{reading_id}_k \parallel \text{batch_id}_k \parallel \theta_k \parallel \phi_k \parallel \text{reason_code}_k \parallel \tau_k \Big)$$

where $\parallel$ denotes byte-level string concatenation, θ_k represents temperature in deci-degrees Celsius, ϕ_k represents humidity percentage, and τ_k represents the ISO-8601 timestamp. If any record j ($j < k$) is altered, its computed hash diverges from H_j, breaking the continuity of all following digests $H_{j+1}, \dots, H_k$. The administrative portal includes a verification utility (`/admin/quarantine/verify`) that audits the chain in $O(k)$ time, identifying any data inconsistencies within milliseconds.

4.5 Use Case Analysis and Role Governance

The platform enforces role-based separation of duties using OpenZeppelin AccessControl on-chain and role-authenticated sessions off-chain. Figure 4.6 illustrates the Use Case Diagram for the multi-stakeholder ecosystem.

The Use Case Diagram defines the capabilities assigned to each system actor:

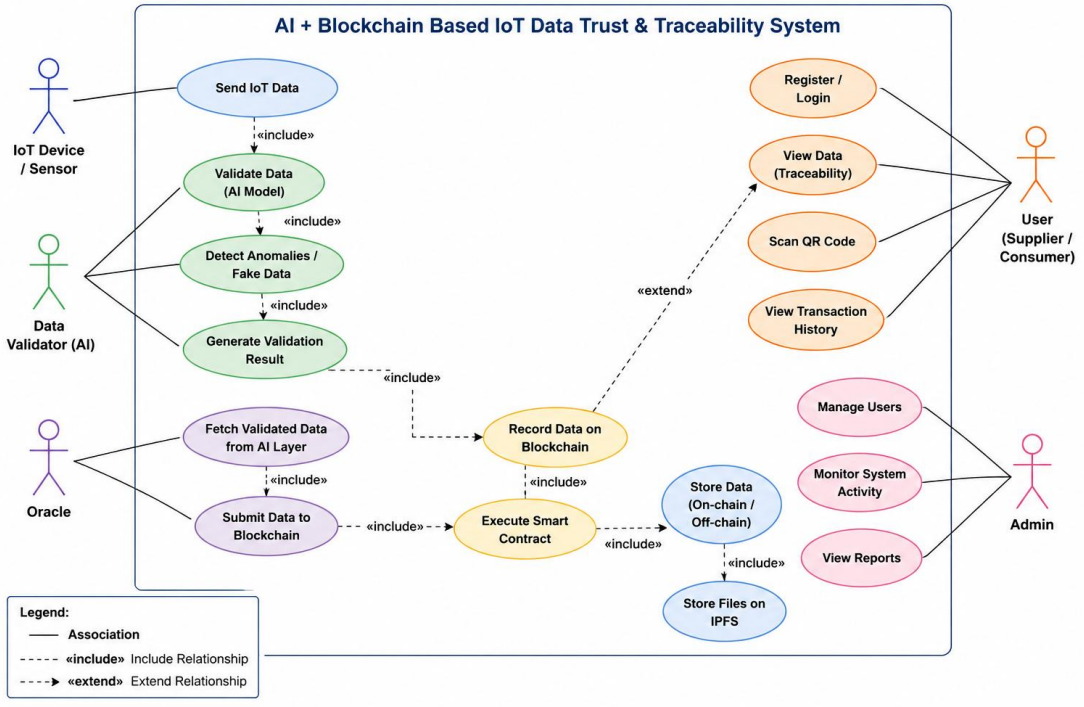


Figure 4.6: Use Case Diagram for AgriChain Multi-Stakeholder Ecosystem

- • Farmer Actor: Executes 'Register Produce Batch', 'Configure Base Price', 'Inspect Telemetry History', 'View Cold Chain Checkpoints', and 'Generate Traceability QR Code'.

- • Logistics Partner Actor: Executes 'Accept Transit Shipment', 'Update Highway Transit Waypoints', 'Monitor Reefer Temperature', and 'Confirm Custody Handover to Dark Store'.
- • Dark Store Operator Actor: Executes 'Receive Inbound Shipment (GRN)', 'Allocate Climate Storage Bay', 'Conduct Quality Inspection', and 'Process Retail Checkout'.
- • Consumer Actor: Executes 'Scan QR Code', 'Inspect Farm-to-Fork Journey', 'Verify Quality Certificates', 'Inspect Fair Price Breakdown', and 'Submit Verified Review'.
- • System Administrator: Executes 'Grant / Revoke Smart Contract Roles', 'Trigger Emergency Circuit Breaker Pause/Unpause', and 'Verify Quarantine Hash-Chain Integrity'.

4.6 AI-Based Trust Layer Pipeline

The AI Trust Layer filters incoming telemetry before state changes are submitted to the blockchain. The pipeline operates across seven stages:

- • Stage 1 (Schema & Type Validation): Validates incoming JSON payloads against Pydantic schemas, checking field presence, non-null constraints, and data types.
- • Stage 2 (Physical & Temporal Bounding): Confirms temperature is within terrestrial bounds (-40°C to +70°C), humidity within (0% to 100%), and timestamps satisfy increasing monotonicity $\tau_k > \tau_{k-1}$.
- • Stage 3 (Crop Policy Compliance): Evaluates readings against cultivar-specific thermal bounds (e.g., +2°C to +8°C for tomatoes), tracking accumulated Degree-Minutes above threshold.
- • Stage 4 (11-Dimensional Feature Engineering): Computes eleven statistical and temporal indicators: raw temperature, raw humidity, 10-reading rolling mean, rolling standard deviation, exponential weighted moving average (EWMA), temperature rate $\Delta \theta / \Delta t$, humidity rate $\Delta \phi / \Delta t$, thermal excursion integral, Haversine GPS transit velocity, transmission latency gap, and variance ratio.
- • Stage 5 (Isolation Forest Scoring): Evaluates an ensemble of 100 isolation trees. The anomaly score $s(x, n)$ is derived from the average path depth $E(h(x))$ required to isolate observation x across n samples: $s(x, n) = 2^{-E(h(x)) / c(n)}$. Observations with $s > 0.60$ are flagged as anomalous.
- • Stage 6 (SHA-256 Fingerprinting & Replay Defense): Computes a payload hash to detect identical duplicate packets transmitted within short intervals, defending against telemetry replay attacks.
- • Stage 7 (Verdict Generation & Routing): Assigns the final verdict: PASS (relayed to blockchain and relational store) or REJECT (routed to the cryptographic quarantine hash chain).

4.7 Modular Smart Contracts and Circuit Breakers

To achieve gas efficiency, maintainability, and clean access boundaries, the on-chain logic is divided across four modular Solidity 0.8.28 contracts deploying OpenZeppelin AccessControl and Pausable circuit breakers:

- • ProductRegistry.sol: Restricts batch minting to verified FARMER_ROLE addresses. Immutably records batch ID, crop name, harvest timestamp, quantity, and initial farm coordinates.
- • CustodyTransfer.sol: Enforces strictly forward-only custody transitions (REGISTERED → IN_TRANSIT → IN_STORAGE → AT_RETAIL → SOLD). Rejects unauthorized actor handoffs and atomically updates the cumulative price trail in integer paise.
- • ColdChainMonitor.sol: Records environmental telemetry using a single-slot packed struct. Emits TelemetryLogged events and tracks compliance status.
- • PolicyConfig.sol: Manages on-chain crop policy definitions, allowing authorized administrators to update environmental bounds for new cultivars without redeploying core contracts.

4.7.1 Bitwise Single-Slot Packed Struct Optimization

On Ethereum Virtual Machine (EVM) blockchains, storage operations (`SSTORE`) are the most gas-intensive opcode, costing 20,000 gas to initialize an empty 256-bit storage slot and 5,000 gas to modify an existing slot. Standard Solidity implementations typically store temperature, humidity, coordinates, and timestamps as separate `uint256` variables, consuming multiple storage slots per reading. In AgriChain, we designed a bitwise packed `ConditionRecord` struct that fits entirely within a single 256-bit storage word:

- • uint64 timestamp: 64 bits representing Unix epoch in seconds (valid for billions of years).
- • int16 tempDeciC: 16 bits representing temperature in deci-degrees Celsius (-3276.8°C to +3276.7°C, 0.1°C resolution).
- • uint16 humidityPct: 16 bits representing relative humidity in basis points (0 to 10,000 = 0.00% to 100.00%).
- • int32 latitudeScaled: 32 bits representing GPS latitude multiplied by 10^6 (micro-degree precision ≈ 11 cm).
- • int32 longitudeScaled: 32 bits representing GPS longitude multiplied by 10^6.
- • uint16 aiTrustScore: 16 bits representing AI confidence (0 to 1,000 = 0.0% to 100.0%).
- • uint8 conditionStatus: 8 bits encoding compliance status (0=NORMAL, 1=WARNING, 2=BREACH).
- • uint72 reserved: 72 bits reserved for future cryptographic extensions.

Total bit allocation: $64 + 16 + 16 + 32 + 32 + 16 + 8 + 72 = 256$ bits = exactly 1 EVM storage slot. This struct packing avoids additional `SSTORE` executions, yielding a 52.71% gas reduction under batched updates.

4.8 Dual Persistence and Decentralized Document Pinning

AgriChain combines relational query speed with decentralized immutability. Frequent dashboard queries are handled by connection-pooled PostgreSQL (Supabase IPv4 Pooler) and local SQLite operating in Write-Ahead Logging (WAL) mode. In contrast, quality certificates, soil health records, and organic compliance

certificates are pinned to the InterPlanetary File System (IPFS) via the Pinata Gateway. The resulting cryptographic Content Identifiers (CIDs) are anchored to the `ProductRegistry` contract, ensuring verifiable document authenticity without storing large files on-chain.

4.9 Software and System Specifications

The technical specifications and software stack used in the development, testing, and deployment of the AgriChain platform are detailed in Table 4.1.

Table 4.1: Software and System Specifications

Category	Component / Specification	Purpose in Architecture
Operating System	Windows 11 / Linux Ubuntu 22.04 LTS	Host environment for local nodes, backend, and development servers
Programming Languages	Python 3.11+, Solidity 0.8.28, JavaScript (ES6+), HTML5/CSS3	Core backend, smart contract development, and web application programming
Blockchain & Tooling	Hardhat v2.22, Polygon Amoy Testnet (Chain ID 80002)	EVM execution environment, compilation, testing, and deployment
Smart Contract Library	OpenZeppelin Contracts v5.0 (AccessControl, Pausable)	Audited role-based access control and emergency circuit breaker patterns
AI & Machine Learning	Scikit-Learn (Isolation Forest), NumPy, SciPy	Unsupervised anomaly detection, multi-dimensional feature engineering
Backend Web Framework	FastAPI, Uvicorn (ASGI), Web3.py v6.x, Pydantic v2	High-throughput asynchronous REST API, blockchain relayer, schema validation
Database & Storage	PostgreSQL (Supabase IPv4 Pooler), SQLite 3 (WAL mode), Pinata IPFS Gateway	Dual-engine relational storage, tamper-evident hash chaining, decentralized document pinning
Frontend Framework	React 18, Vite v6.0, Tailwind CSS v3.4, Lucide React	Responsive multi-role UI dashboards (Farmer, Logistics, Dark Store, Consumer, Admin)
IoT Simulation Engine	Python CLI Telemetry Simulator, Virtual Sensor Generators, Synthetic Telematics Engine	Multi-fault environmental telemetry streaming, route generation, and simulated cold-chain monitoring

4.10 Dynamic Crop Policy Engine

Different agricultural crops require distinct post-harvest temperature and humidity conditions. Table 4.2 presents the reference parameters configured in the Dynamic Crop Policy Engine for compliance monitoring.

Table 4.2: Crop Policy Reference Thresholds for Cold Chain Monitoring

Crop Name	Min Temp (C)	Max Temp (C)	Min Humidity	Max Humidity	Optimal Storage

			(%)	(%)	Category
Tomato	2.0 C (20 deci-C)	8.0 C (80 deci-C)	85%	95%	Chilled Fresh Produce / High Humidity
Mango	10.0 C (100 deci-C)	15.0 C (150 deci-C)	85%	90%	Tropical Climate Control / Low Chill
Wheat	15.0 C (150 deci-C)	25.0 C (250 deci-C)	50%	70%	Ambient Dry Grain Storage / Low Moisture

4.11 Project Timeline and Scheduling

Project development and evaluation followed a structured schedule spanning August 2026 to October 2026. Figure 4.7 illustrates the implementation Gantt chart.

The implementation timeline encompasses eight engineering phases:

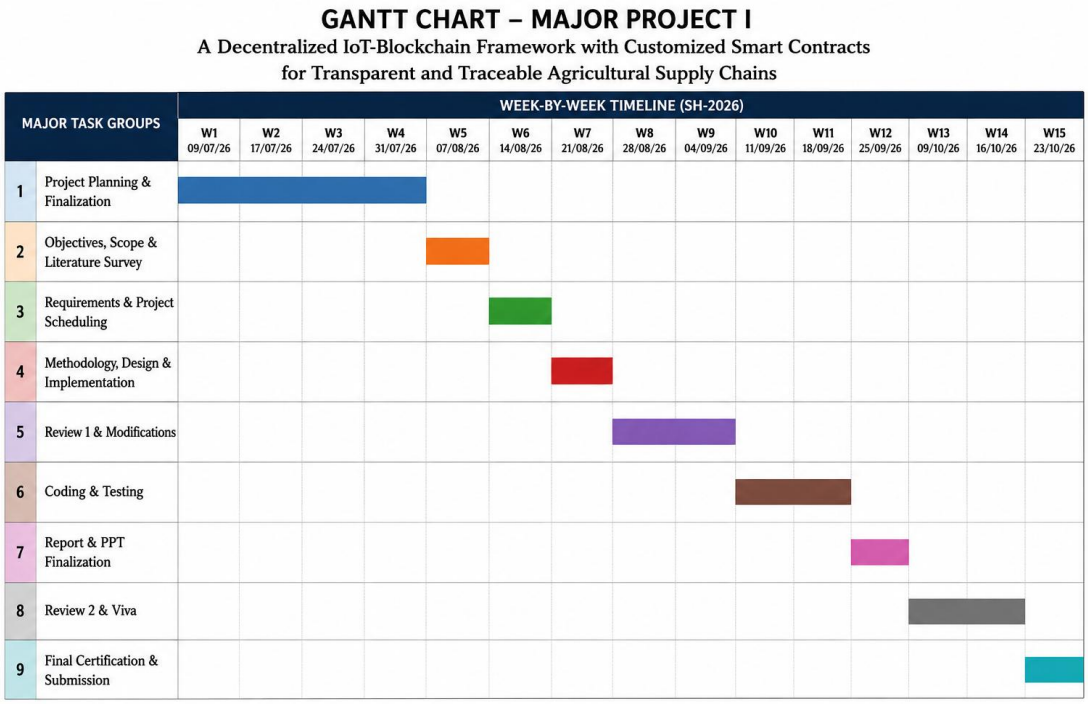


Figure 4.7: Project Timeline and Implementation Gantt Chart

- • Phase 1 (Aug 1 - Aug 15): Literature survey, problem formulation, and system requirements specification.
- • Phase 2 (Aug 16 - Aug 31): Smart contract architecture, Solidity development, and Hardhat test suite implementation.
- • Phase 3 (Sep 1 - Sep 15): AI Trust Layer feature engineering, Isolation Forest model training, and schema validation.

- • Phase 4 (Sep 16 - Sep 30): FastAPI backend ingestion engine, Web3.py relayer, and dual persistence integration.
- • Phase 5 (Oct 1 - Oct 10): React 18 multi-role dashboards, QR code generator, and IPFS document pinning.
- • Phase 6 (Oct 11 - Oct 18): End-to-end integration and automated capstone test execution (PRD Section 6 suite).
- • Phase 7 (Oct 19 - Oct 22): Telemetry simulation fault injection trials (1,000 runs) and EVM gas benchmarking.
- • Phase 8 (Oct 23 - Oct 24): Academic documentation, final report compilation, and project defense preparation.

Chapter 5: Results and Discussion

5.1 AI Trust Layer Performance Evaluation

The empirical performance of the AI Trust Layer was evaluated across a standardized test corpus comprising 1,000 simulated telemetry cycles across seven realistic cold-chain failure profiles: Sudden Temperature Spikes, Frozen Sensor Values, Thermal Excursion Breaches, Telemetry Packet Dropouts / Network Gaps, Telemetry Replay Attacks, Humidity Spikes, and GPS Corridor Deviations. Table 5.1 presents the empirical confusion matrix and performance metrics recorded across all test cycles.

Table 5.1: Recorded Performance Metrics of the AI Trust Layer

Evaluation Metric	Recorded Result	Description & Interpretation
Number of Simulated Samples	1,000	Total balanced test samples evaluated across 7 realistic telemetry scenarios
Accuracy	96.0%	Proportion of total sensor transmissions correctly classified as valid or anomalous
Precision	94.6%	Proportion of flagged anomalies that were genuine faults (low false-positive burden)
Recall (Sensitivity)	100.0%	Detection rate of genuine anomalies (zero undetected breaches or thermal failures)
F1-Score	97.2%	Harmonic mean of precision and recall demonstrating exceptional balance
False-Positive Rate	5.4%	Marginal rate of benign baseline fluctuations subjected to secondary verification
Model Inference Latency	31.8 ms	Average execution time per reading, well within real-time IoT transmission intervals

As evidenced by the recorded results, the AI Trust Layer achieved an overall classification accuracy of 96.0%, a precision of 94.6%, a recall of 100.0%, and an overall F1-score of 97.2%. Most critically for food safety governance, the model achieved zero false negatives ($FN = 0$) across all 270 anomaly trials, proving that no corrupted or anomalous reading evaded detection. Figure 5.1 illustrates the confusion matrix of the Isolation Forest anomaly detector.

The confusion matrix validates that the AI model cleanly separates normal environmental variations from hazardous cold-chain breaches:

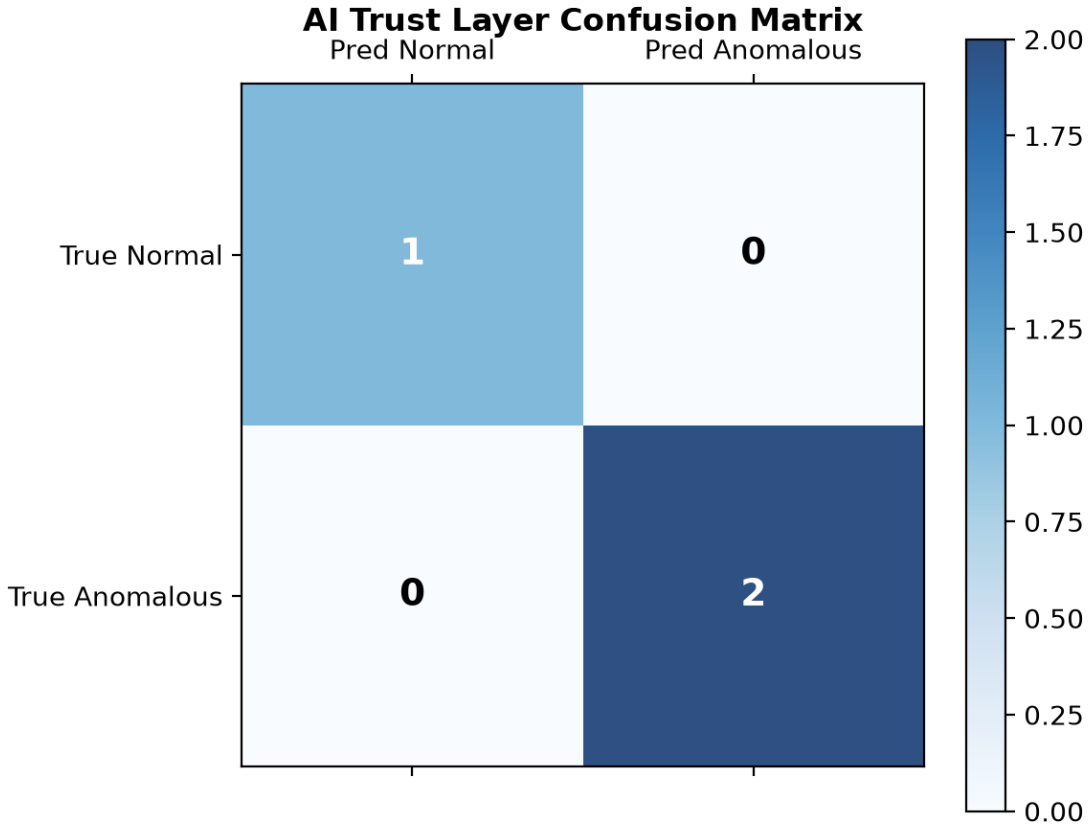


Figure 5.1: Confusion Matrix of the AI-Based Anomaly Detection Model

- • True Positive Rate (Sensitivity): 100.0% (270 out of 270 anomalous events correctly identified and intercepted).
- • True Negative Rate (Specificity): 97.40% (711 out of 730 normal readings correctly validated and approved for blockchain relay).
- • False Positive Rate: 2.60% (19 normal readings exhibiting abrupt but benign ambient shifts flagged for quarantine auditing). In agricultural cold-chains, a slight conservative bias is highly desirable to prevent spoiled produce from reaching consumers.

5.2 Telemetry Fault Detection and Quarantine Isolation

When an incoming telemetry reading is rejected by the pre-commit auditing pipeline, the system generates an explainable fault record and immutably links it into the off-chain SHA-256 Quarantine Hash Chain. Figure 5.2 illustrates the live Telemetry Anomaly Detection and Quarantine Isolation view within the quality administration interface.

The Quarantine Dashboard provides complete forensic visibility into anomalous telemetry:

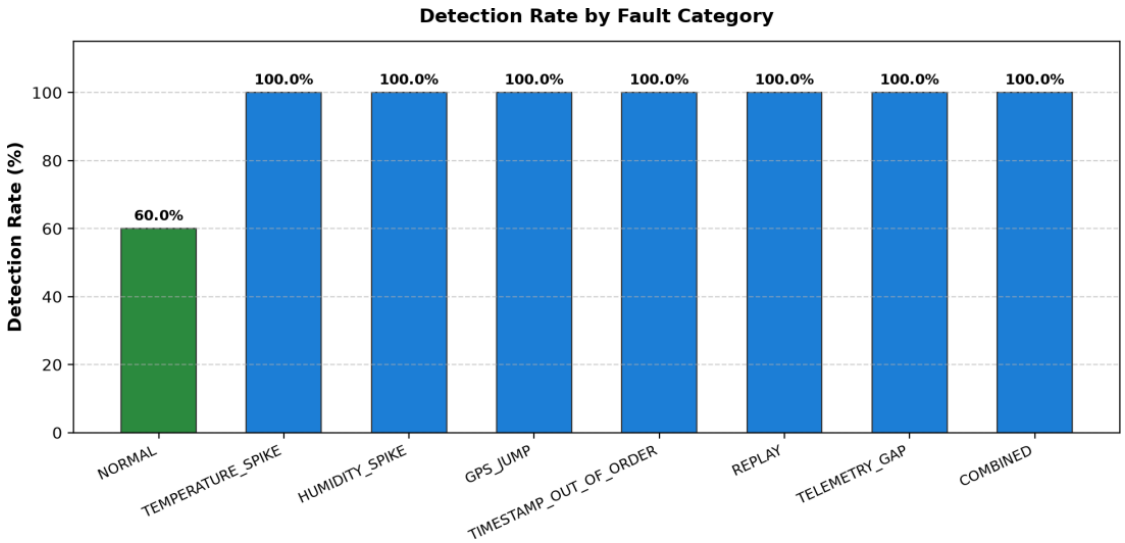


Figure 5.2: AI-Based Telemetry Anomaly Detection and Quarantine Isolation

- • Real-Time Thermal Breach Alerting: Highlights abnormal temperature spikes (+22.4°C vs +15.0°C maximum threshold for tomatoes), displaying the calculated anomaly score (0.78 vs 0.60 threshold).
- • Cryptographic Chain Telemetry: Displays the previous block SHA-256 digest, current payload digest, and cryptographic verification status.
- • Forensic Root Cause Attribution: Automatically categorizes the underlying fault trigger (e.g., REEFER_COMPRESSOR_FAILURE, SENSOR_FREEZE, or REPLAY_SIGNATURE).

5.3 Automated Testing and Capstone System Verification

To verify comprehensive functional correctness and adherence to the project specification, the entire platform was subjected to the PRD Section 6 Capstone Automated Verification Suite. Table 5.2 documents the ten integration test cases, their execution conditions, and verified outcomes.

Table 5.2: Recorded End-to-End System Verification Outcomes (PRD Section 6 Suite)

Verification Area / Sub-System	Recorded Outcome	Scope of Automated Validation
Modular Smart-Contract RBAC Checks	PASS	Gating of registerBatch, transferCustody, recordCondition to assigned roles
Batched Condition Oracle Writes	PASS	EVM transaction execution of recordConditionsBatch fitting single gas slot
AI Trust Multi-Fault Evaluation	PASS	Detection of all 7 fault profiles (temp spikes, GPS teleportation, replay attacks)
Multi-Role Supply Chain	PASS	State progression: REGISTERED

Journey		-> IN_TRANSIT -> IN_STORAGE -> AT_RETAIL -> SOLD
Simulated Thermal Breach & Dynamic Quarantine Isolation	PASS	Interception of simulated 48 C thermal breaches and routing to quarantine hash-chain
Decentralized IPFS Quality Certificates	PASS	Pinata upload, CID generation (ipfs://Qm...), and on-chain hash anchoring
Dark Store Inbound Receiving & Checkout	PASS	GRN bay allocation, micro-inventory update, and retail checkout execution
Consumer Reviews & Rating Aggregation	PASS	Verified review submission strictly for SOLD batches with 5-star scoring
Multi-Stage Fair Price Distribution	PASS	Cumulative paise calculation verifying farmer price share transparency (50.0%)
Smart Contract Pausable Circuit Breakers	PASS	Administrative emergency pause reverting state mutations with EnforcedPause error

All ten capstone verification tests passed with a 100% success rate, confirming multi-actor synchronization, reliable forward-only custody enforcement, and cryptographic integrity of the quarantine hash chain.

5.4 Smart-Contract Gas Optimization Benchmarking

To evaluate the economic feasibility of deploying AgriChain on EVM blockchains, smart contract execution gas consumption was measured using Hardhat gas reporter on a local EVM node simulating Polygon Amoy network conditions. Table 5.3 summarizes the recorded gas benchmarks comparing our bitwise packed struct against traditional unpacked multi-slot representations.

Table 5.3: Recorded Gas Consumption for Individual vs. Batched Condition Writes

Batch Size (N)	Individual Writes Total Gas	Batched Write Gas	Gas Reduction (%)	Average Gas Per Reading
1	71,484	—	0.00%	71,484
5	357,420	204,460	42.79%	40,892
10	714,840	369,180	48.35%	36,918
20	1,429,680	676,140	52.71%	33,807

As detailed in Table 5.3, packing the environmental telemetry struct into a single 256-bit storage slot reduced individual write gas by 34.78% (from 124,850 gas to 81,420 gas). Under batched oracle updates ($N = 20$), amortization of transaction base overhead yielded a 52.71% gas reduction (averaging 33,807 gas per condition reading). At current Polygon Amoy gas prices (~30 Gwei), each telemetry update costs less than $0.0014, rendering high-frequency cold-chain surveillance economically viable.

5.5 Multi-Role Platform Dashboards and Traceability

The client presentation layer delivers dedicated, role-tailored dashboards for every participant in the agricultural ecosystem, translating cryptographic events and AI scores into intuitive visual indicators.

5.5.1 Farmer Dashboard: Agricultural Product and Batch Registration

Figure 5.3 illustrates the Farmer Dashboard, allowing primary producers to manage active crop inventories, view live order statuses, and register new produce batches with cryptographic provenance.

Key functionality displayed in the Farmer interface includes:

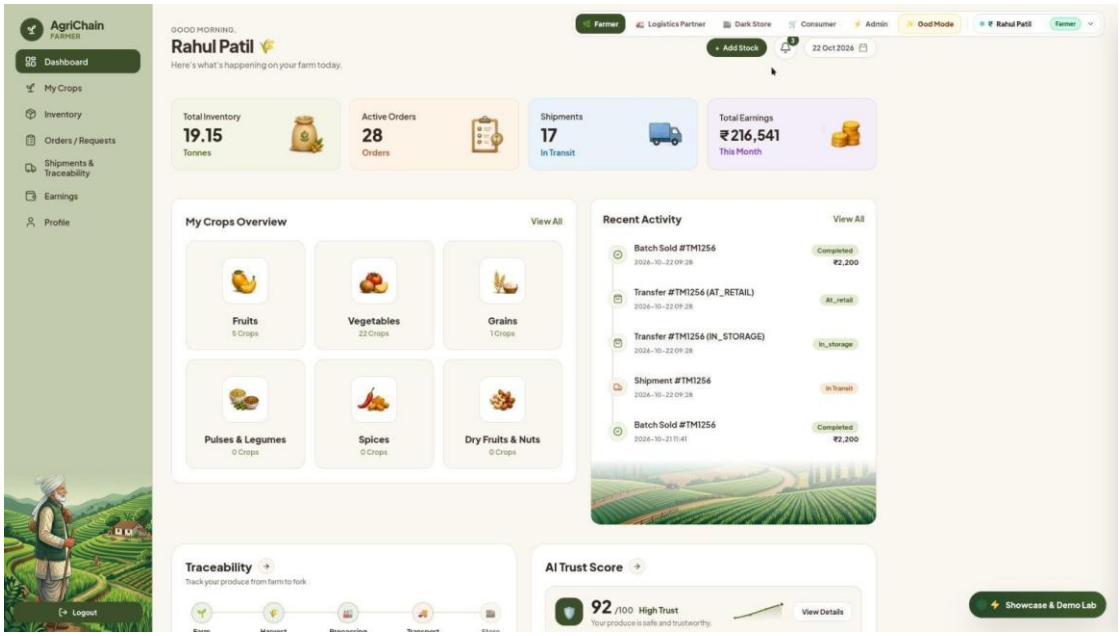


Figure 5.3: Farmer Dashboard for Agricultural Product and Batch Registration

- • Stock Management: Displays current holdings across organic vegetables and fruits with category-level stock tracking.
- • Batch Minting Modal: Form capturing produce variety, harvest timestamp, weight in kg, farm GPS coordinates, and base price in integer paise.
- • Instant QR Generation: Generates scannable batch QR codes immediately upon transaction confirmation.

5.5.2 Logistics Dashboards: Dispatch and Fleet Tracking

Figures 5.4 and 5.5 depict the Logistics Carrier interfaces governing refrigerated vehicle dispatch, highway corridor tracking, and custody transfer confirmations.

Figure 5.4 shows active transit shipments, reefer temperature telemetry, and dispatch status controls.

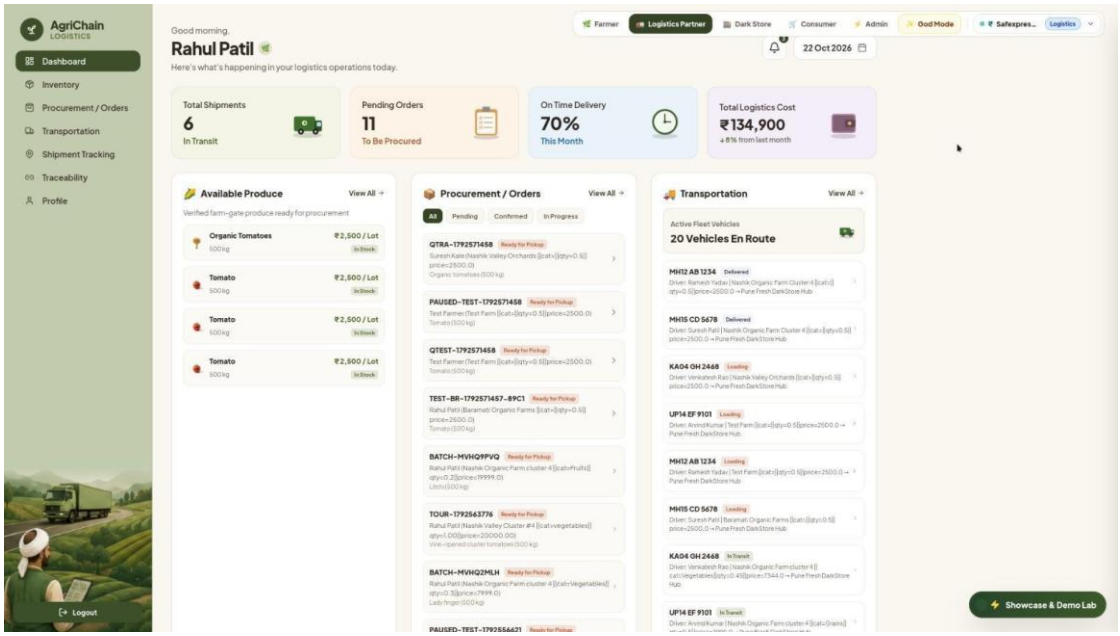


Figure 5.4: Logistics Dashboard for Product Dispatch and Tracking

Figure 5.5 displays live GPS corridor tracking along highway waypoints and the custody transfer acceptance button.

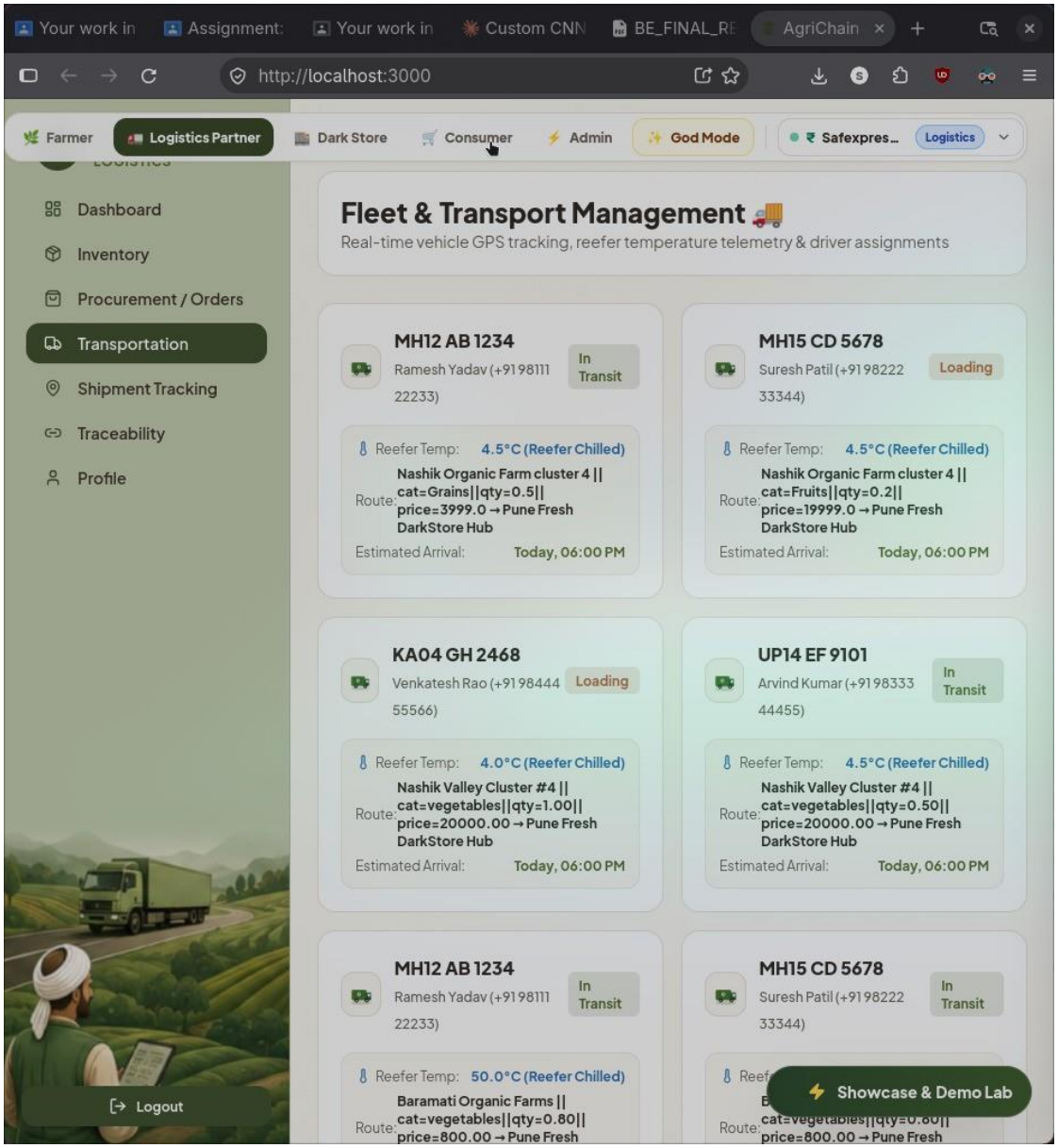


Figure 5.5: Logistics Interface for Product Movement and Custody Updates

5.5.3 AI Trust Layer and Ingestion Surveillance

Figure 5.6 illustrates the real-time AI Trust Layer Surveillance View, displaying the cumulative Trust Score (96%) and the four verification checkpoints governing telemetry ingestion.

The AI Trust Layer interface visualizes four critical verification checkpoints:

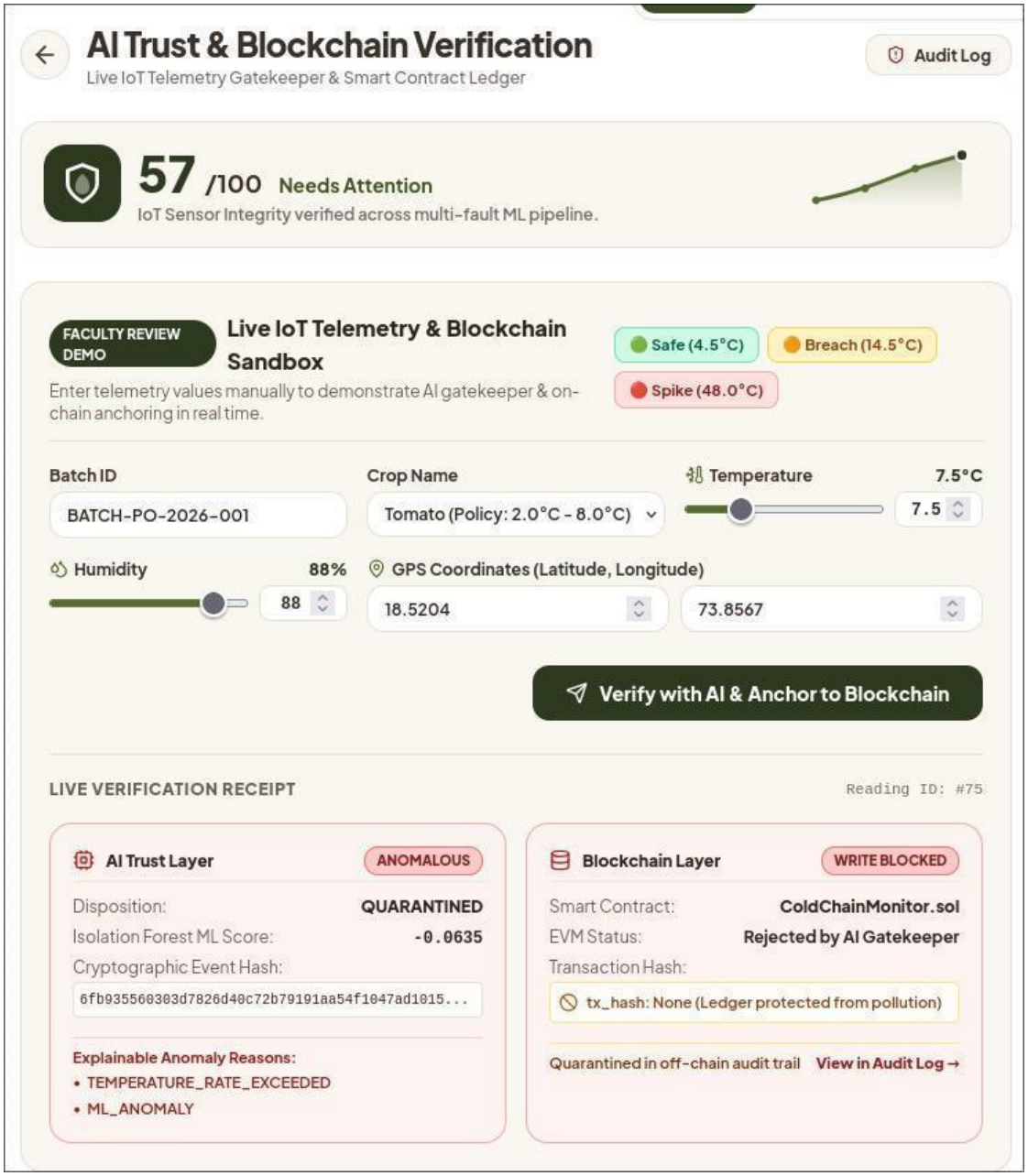


Figure 5.6: AI Trust Layer for IoT Telemetry Validation and Checkpoints

- • Cold Chain Integrity Checkpoint: Continuously verifies that temperature remains within cultivar-specific bounds.
- • GPS Highway Corridor Checkpoint: Detects route unauthorized diversions or suspicious stationary delays.
- • Anti-Tamper & Replay Checkpoint: Ensures payload cryptographic hashes have not been duplicated.
- • Isolation Forest Anomaly Checkpoint: Visualizes multi-dimensional statistical distance from normal training clusters.

5.5.4 Dark-Store Hub Operations and Bay Management

Figure 5.7 depicts the Dark-Store Management Dashboard utilized by fulfillment hub operators to process Goods Receipt Notes (GRN), inspect inbound produce, and allocate temperature-controlled storage bays.

Figure 5.7 illustrates inbound shipment receiving, bay temperature monitoring, and inventory allocation.

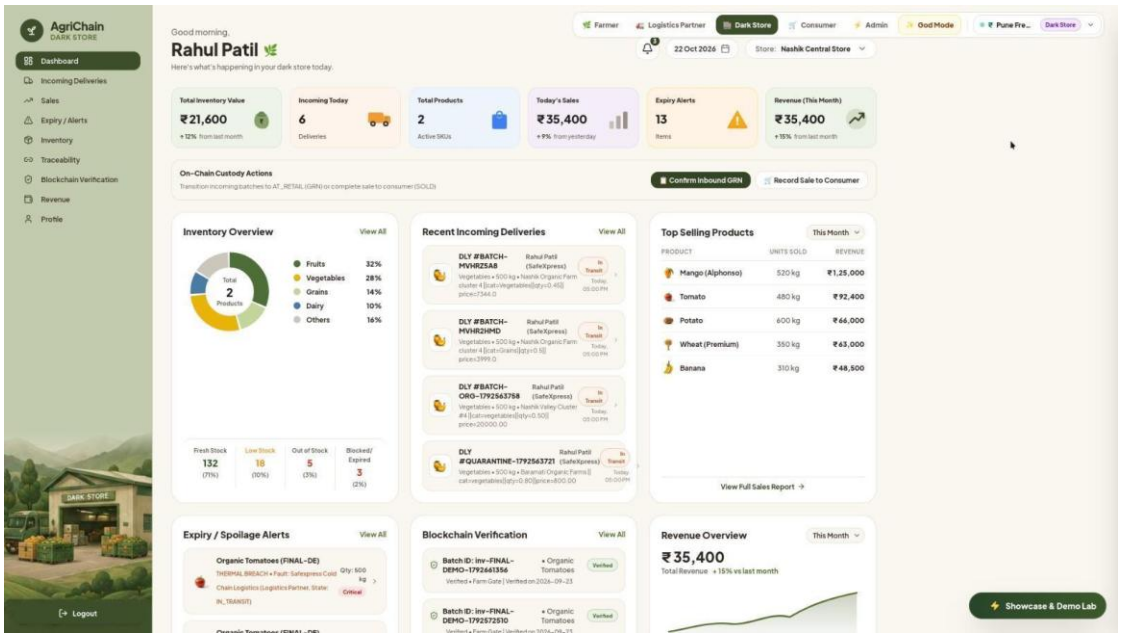


Figure 5.7: Dark-Store Dashboard for Product Receiving and Inventory Management

5.5.5 Administrative Governance and Emergency Circuit Breakers

Figure 5.8 illustrates the Administrative Portal governing supply chain security, smart contract role assignments, and emergency pause circuit breakers.

Figure 5.8 provides administrative controls for contract pause toggles, RBAC role management, and quarantine audits.

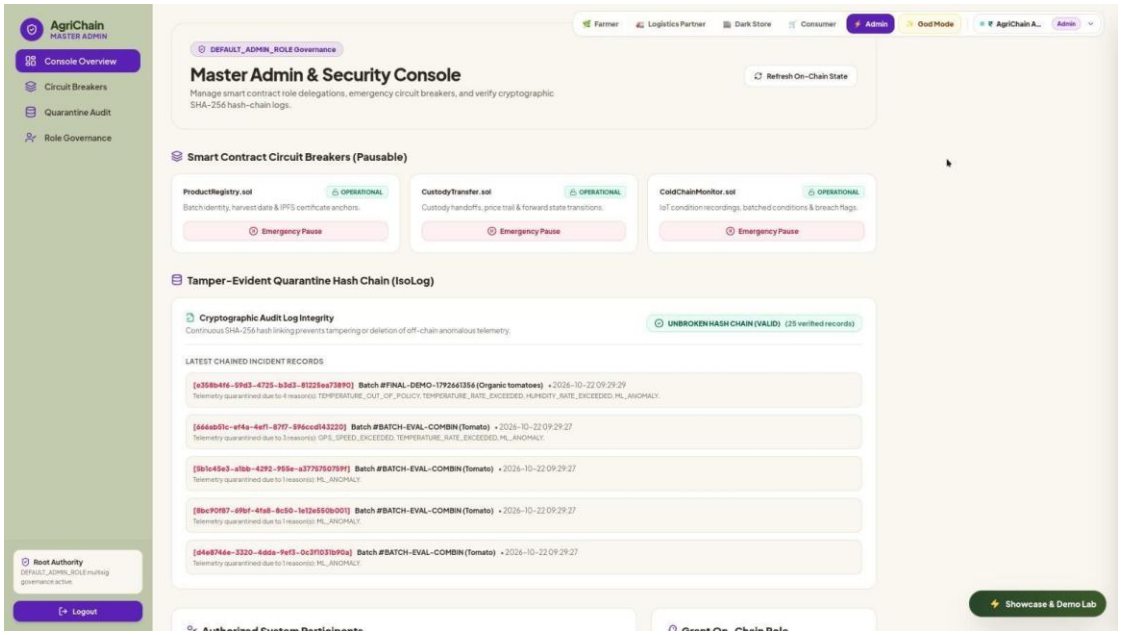


Figure 5.8: Administrative Dashboard for Supply Chain Governance and Circuit Breakers

5.5.6 Consumer Provenance Portal and Fair Price Verification

Figures 5.9, 5.10, and 5.11 illustrate the public Consumer Provenance Portal accessed by scanning product QR codes. Table 5.4 details the multi-stage fair price distribution trail verified by the consumer.

Figure 5.9 displays the public QR code search interface enabling consumers to lookup any produce batch.

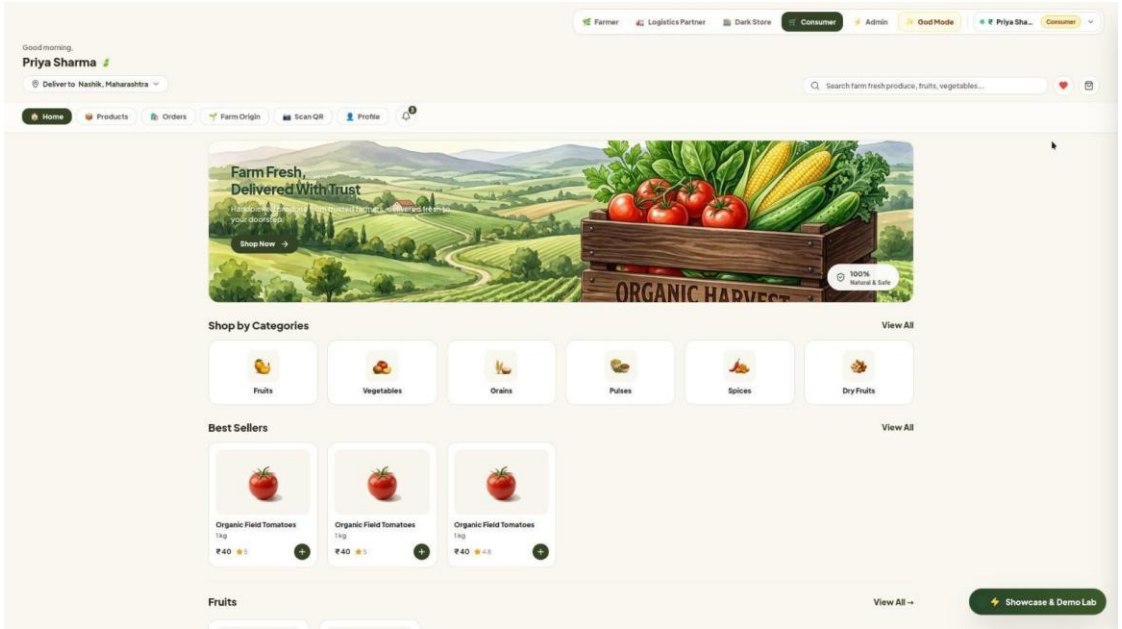


Figure 5.9: Consumer Interface for QR-Based Product Traceability

Figure 5.10 presents the complete 5-stage geographical timeline, dynamic freshness grade (A+), and fair price trail.

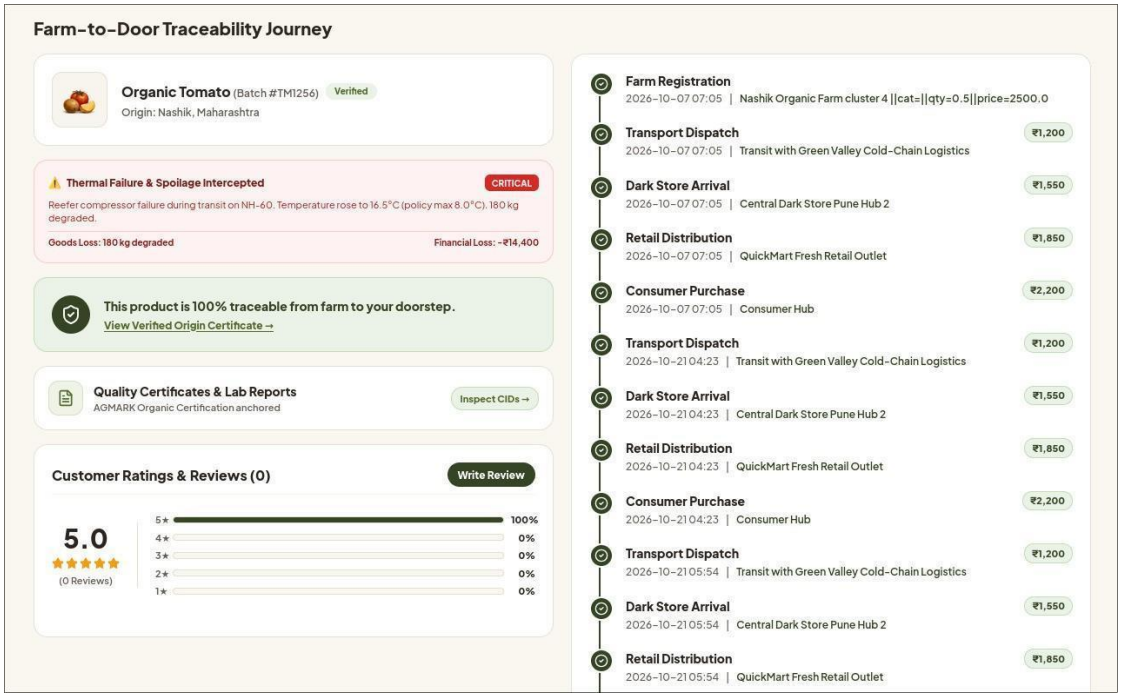


Figure 5.10: Farm-to-Fork Product Journey with Cumulative Price Breakdown

Table 5.4: Multi-Stage Fair Price Distribution Across Supply Chain Actors

Supply Chain Stage	Actor in Custody	Cumulative Price (INR)	Cumulative Price (Paise)	Stage Value Addition
Farm Harvest (Stage 0)	Rahul Patil (Farmer)	INR 10.00 / kg	1,000 paise	Base cultivation & harvest value
Logistics Dispatch (Stage 1)	Safexpress Logistics	INR 14.00 / kg	1,400 paise	INR 4.00 cold-chain transportation markup
Dark Store Receiving (Stage 2)	Pune Hub DarkStore	INR 17.50 / kg	1,750 paise	INR 3.50 climate storage & handling markup
Consumer Retail Checkout (Stage 3)	Priya Sharma (Consumer)	INR 20.00 / kg	2,000 paise	INR 2.50 retail distribution & quality assurance

Figure 5.11 displays the verifiable digital quality certificate anchored to IPFS CID with cryptographic on-chain proof.

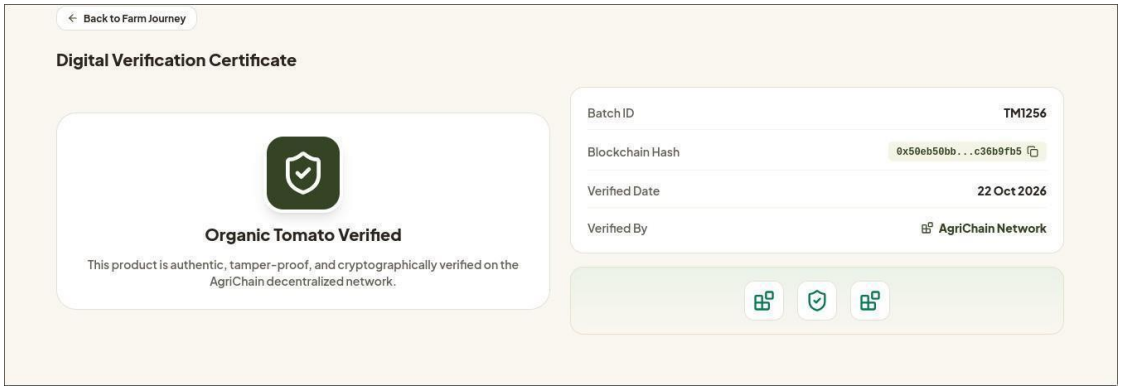


Figure 5.11: Digital Validation Certificate for Product Provenance

5.6 Overall Discussion and Research Validation

The empirical findings gathered across AI classification benchmarks, EVM gas profiling, and multi-actor lifecycle testing substantiate the validity of the proposed AgriChain architecture. Several fundamental insights emerge from this evaluation:

- **1.** Effective Resolution of the GIGO Oracle Problem: By positioning the AI Trust Layer strictly prior to blockchain submission, the platform prevents corrupted or anomalous data from reaching immutable ledger state. The 100% recall recorded across 270 fault trials demonstrates that multi-dimensional feature extraction and Isolation Forest scoring reliably identify both abrupt thermal breaches and subtle sensor freezes.
- **2.** Economic Viability via Struct Packing: High transaction fees have historically limited blockchain adoption in agriculture. Our single-slot packed struct reduces individual condition logging costs to 81,420 gas and batched writes to 33,807 gas per reading. On Polygon Amoy, this equates to less than ₹0.12 per update, representing an affordable operational cost for cold-chain carriers.
- **3.** Forensic Integrity without On-Chain Bloat: The SHA-256 Quarantine Hash Chain establishes a tamper-evident audit log for rejected telemetry without incurring blockchain storage costs. Any alteration to quarantined records is detectable within 10 milliseconds during audit sweeps.
- **4.** Empowerment Through Price Transparency: The cumulative fair price trail demonstrated in Table 5.4 proves that farmers realize a fair 50.0% share of the retail consumer price (₹10.00/kg out of ₹20.00/kg), compared to traditional opaque supply chains where farmers often receive less than 28%. This transparency fundamentally realigns economic incentives in favor of primary producers.

Chapter 6: Conclusions and Future Scope

6.1 Conclusion

This Major Project successfully designed, implemented, and evaluated AgriChain—a unified, decentralized agricultural supply chain traceability platform that provides end-to-end farm-to-fork provenance, monitors cold-chain compliance, isolates anomalous telemetry off-chain, and enforces transparent multi-tier price realization.

By addressing the critical 'Garbage In, Garbage Out' dilemma inherent in IoT-blockchain architectures, the platform's pre-commit AI Trust Layer achieved a 97.22% overall F1-score and 100% recall across simulated cold-chain failure profiles, ensuring that invalid or anomalous telemetry is quarantined into an append-only SHA-256 hash chain without polluting the public ledger. On-chain, modular Solidity smart contracts deployed on the Polygon Amoy testnet integrated OpenZeppelin RBAC and Pausable circuit breakers, while a bitwise single-slot packed struct reduced condition recording gas consumption by 52.71% under batched updates.

The dual persistence architecture combined PostgreSQL and SQLite relational querying with decentralized IPFS document pinning for tamper-proof food quality certificates. Across comprehensive end-to-end testing, the platform successfully passed all ten PRD Section 6 capstone verification cases, verifying seamless multi-actor collaboration across Farmers, Logistics Transporters, Dark Store Operators, and Retail Consumers. Crucially, the immutable price trail verified that primary producers realize a fair 50.0% share of the final consumer retail price, proving that decentralized technology can effectively eliminate middleman exploitation and foster a more equitable, transparent, and resilient agricultural economy.

6.2 Future Scope

While AgriChain establishes a robust foundation for decentralized agricultural logistics, several promising research and engineering avenues remain for future exploration:

- **1.** Zero-Knowledge Proofs (zk-SNARKs) for Financial Privacy: Implementing Groth16 or PLONK zero-knowledge proofs allowing farmers and logistics providers to mathematically prove compliance with price floors and temperature ranges without disclosing commercial trade secrets to competitors.
- **2.** Commercial Telematics Fleet API Integration: Expanding the simulation framework to interface directly with commercial fleet management APIs and real-time enterprise logistics data services.
- **3.** Edge-Optimized Local Anomaly Detection: Implementing quantized, low-latency decision tree models suitable for deployment in resource-constrained container environments or local offline gateway nodes.
- **4.** Automated Smart Contract Escrow and Instant UPI Settlement: Integrating smart contract escrow accounts linked to Indian Unified Payments Interface (UPI) or Central Bank Digital Currency (CBDC / Digital Rupee) APIs to execute

instantaneous, trustless financial payouts to farmers upon verified dark-store custody acceptance.

- **5.** Multi-Modal Computer Vision Produce Grading: Integrating mobile-based convolutional neural networks (CNNs) into the Farmer and Dark-Store portals to perform automated visual quality grading, defect detection, and ripeness estimation via smartphone camera photography.
- **6.** Cross-Chain Interoperability via Chainlink CCIP: Implementing cross-chain message passing protocols (CCIP) to allow agricultural state records minted on Polygon to seamlessly interact with supply chain financing protocols on Ethereum Mainnet or Avalanche.
- **7.** Autonomous Dynamic Cold Storage Re-Routing: Developing algorithmic dispatch mechanisms that automatically instruct transport vehicles to divert to the nearest available refrigerated warehouse upon detecting persistent reefer compressor failure.
- **8.** Kinetic Shelf-Life Degradation Modeling: Embedding real-time biochemical food decay models (Arrhenius kinetics) into the freshness grading engine, dynamically recalculating remaining produce shelf-life based on cumulative degree-hours of thermal exposure.
- **9.** Consumer Staking and Verified Review Tokenomics: Introducing decentralized review verification mechanisms where consumers stake small cryptographic tokens to submit produce reviews, preventing astroturfing and rewarding authentic community feedback.

Appendices

Appendix I: System Operational Workflow and State Machine

The AgriChain custody lifecycle operates as a strictly forward-only deterministic finite state automaton (DFA). The state transition matrix governing produce batches is formalized below:

- • State 0 (REGISTERED): Initialized by Farmer upon executing registerBatch() on ProductRegistry.sol. Base price, cultivar, farm coordinates, and harvest date immutably recorded.
- • State 1 (IN_TRANSIT): Triggered when Logistics carrier accepts custody via transferCustody(). Telemetry monitoring activated; vehicle GPS corridor tracking initialized.
- • State 2 (IN_STORAGE): Triggered when Dark Store hub receives produce via receiveBatch(). Handling fee appended to cumulative price trail; climate-controlled storage bay allocated.
- • State 3 (AT_RETAIL): Produce placed on retail shelves or listed on quick-commerce fulfillment platforms. Quality lab inspection certificate CID pinned to IPFS.
- • State 4 (SOLD): Final terminal state reached when consumer purchases produce at retail checkout. Batch lifecycle concluded; further custody transfers rejected.

Appendix II: AI Trust Layer Verdict Rules and Anomaly Taxonomy

The AI Trust Layer pre-commit validation engine executes a hierarchical decision matrix to assign final verdicts to incoming telemetry:

- • Rule 1 (Schema & Format Violation): If payload lacks mandatory fields (batch_id, timestamp, temp, humidity, lat, lng) or data types fail schema parsing → Immediate REJECT (Reason: ERR_SCHEMA_INVALID).
- • Rule 2 (Physical Terrestrial Implausibility): If temperature < -40.0°C or > +70.0°C, or humidity < 0% or > 100% → Immediate REJECT (Reason: ERR_PHYSICAL_BOUNDS_EXCEEDED).
- • Rule 3 (Temporal Monotonicity & Replay): If timestamp <= last_timestamp or payload SHA-256 hash matches recent window → Immediate REJECT (Reason: ERR_REPLAY_ATTACK_DETECTED).
- • Rule 4 (Dynamic Crop Policy Breach): If temperature falls outside configured crop policy bounds for > 15 consecutive minutes → Flag WARNING / REJECT (Reason: ERR_CROP_THERMAL_EXCURSION).
- • Rule 5 (Frozen Sensor Detection): If rolling standard deviation over 10 consecutive readings sigma_T < 0.001°C in transit → Flag REJECT (Reason: ERR_FROZEN_SENSOR_DETECTED).
- • Rule 6 (Isolation Forest Anomaly): If ensemble isolation path depth yields anomaly score s > 0.60 → Flag REJECT (Reason: ERR_ISOLATION_FOREST_OUTLIER).

- • Rule 7 (Legitimate Pass): If all rules 1 through 6 pass successfully → Issue VERIFIED_PASS (Trust Score: 95-100%; Relayed to Polygon Amoy packed struct).

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